

Practical Workbook

CT-376

COMPUTER COMMUNICATION NETWORKS



Name: _____

Roll No: _____

Batch: _____

Department: _____

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Lab # 01

Objective

Introduction to networking and installation of network simulator.

Theory

Data Communication

Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable. For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs). The effectiveness of a data communications system depends on four fundamental characteristics: delivery, accuracy, timeliness, and jitter.

1. **Delivery.** The system must deliver data to the correct destination.
2. **Accuracy.** The system must deliver the data accurately.
3. **Timeliness.** The system must deliver data in a timely manner.
4. **Jitter.** Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets.

Components

1. **Message.** The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
2. **Sender.** The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
3. **Receiver.** The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
4. **Transmission medium.** The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.
5. **Protocol.** A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

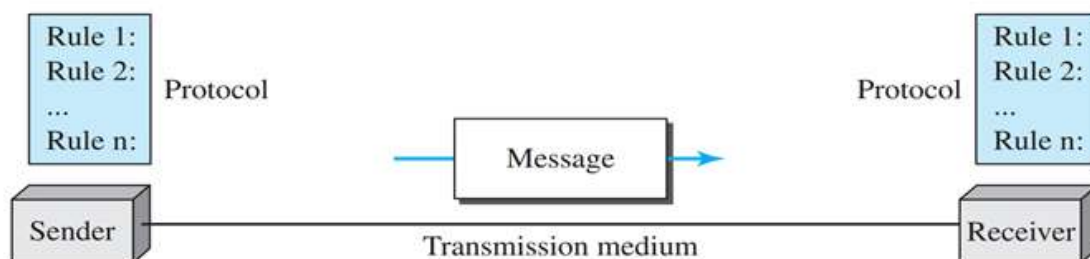


Figure 1 Components of Data Communication

Networks

A network is the interconnection of a set of devices capable of communication. In this definition, a device can be a host (or an end system as it is sometimes called) such as a large computer, desktop, laptop, workstation, cellular phone, or security system. A device in this definition can also be a connecting device such as a router, which connects the network to other networks, a switch, which connects devices together, a modem (modulator-demodulator), which changes the form of data, and so on. These devices in a network are connected using wired or wireless transmission media such as cable or air. When we connect two computers at home using a plug-and-play router, we have created a network, although very small.

Types of Networks

- **Local Area Network:** A local area network (LAN) is usually privately owned and connects some hosts in a single office, building, or campus. Each host in a LAN has an identifier, an address, that uniquely defines the host in the LAN. A packet sent by a host to another host carries both the source host's and the destination host's addresses.
- **Wide Area Network:** A wide area network (WAN) is also an interconnection of devices capable of communication. A WAN has a wider geographical span, spanning a town, a state, a country, or even the world. A WAN is normally created and run by communication companies and leased by an organization that uses it. We see two distinct examples of WANs today: point-to-point WANs and switched WANs.

Switching

An internet is a switched network in which a switch connects at least two links together. A switch needs to forward data from a network to another network when required. The two most common types of switched networks are circuit-switched and packet-switched networks.

Circuit-Switched Network

In a circuit-switched network, a dedicated connection, called a circuit, is always available between the two end systems; the switch can only make it active or inactive. Figure 2 shows a very simple switched network that connects four telephones to each end. We have used telephone sets instead of computers as an end system because circuit switching was very common in telephone networks in the past, although part of the telephone network today is a packet-switched network.

Packet-Switched Network

In a computer network, the communication between the two ends is done in blocks of data called packets. This allows us to make the switches function for both storing and forwarding because a packet is an independent entity that can be stored and sent later. Figure 3 shows a small packet-switched network that connects four computers at one site to four computers at the other site.

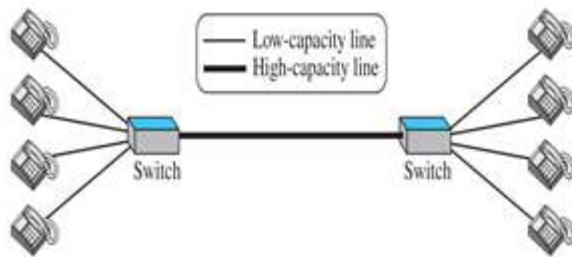


Figure 2 Circuit-Switched Network

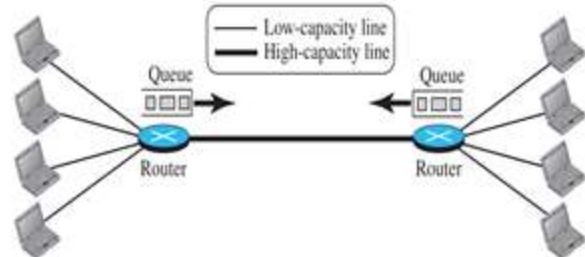


Figure 3 Packet-Switched Network

THE OSI MODEL

An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late 1970s. An **open system** is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture. The purpose of the OSI model is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software. The OSI model is not a protocol; it is a model for understanding and designing a network architecture that is flexible, robust, and interoperable. The OSI model was intended to be the basis for the creation of the protocols in the OSI stack.

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium

Figure 4 OSI Model

TCP/IP PROTOCOL SUITE

TCP/IP is a protocol suite (a set of protocols organized in different layers) used in the Internet today. It is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality. The term hierarchical means that each upper-level protocol is supported by the services provided by one or more lower-level protocols. The original TCP/IP protocol suite was defined as four software layers built upon the hardware. Today, however, TCP/IP is thought of as a five-layer model.

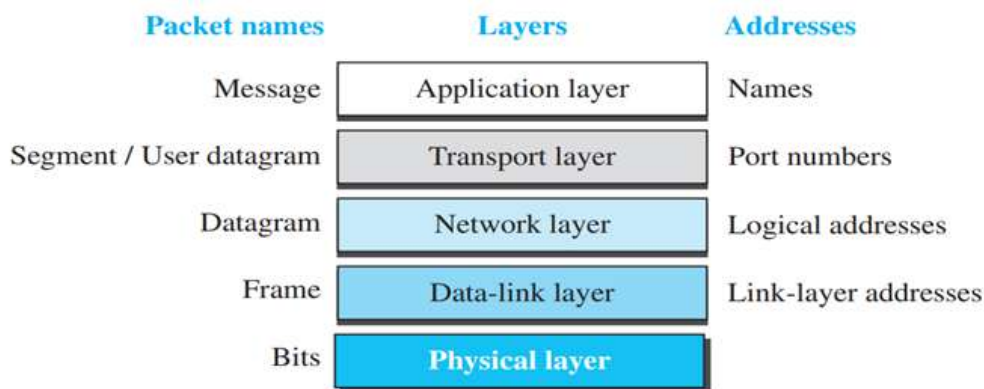


Figure 5 TCP/IP Model

Using logical connections makes it easier for us to think about the duty of each layer. The domain of duty of the top three layers is the internet, and the domain of duty of the two lower layers is the link.

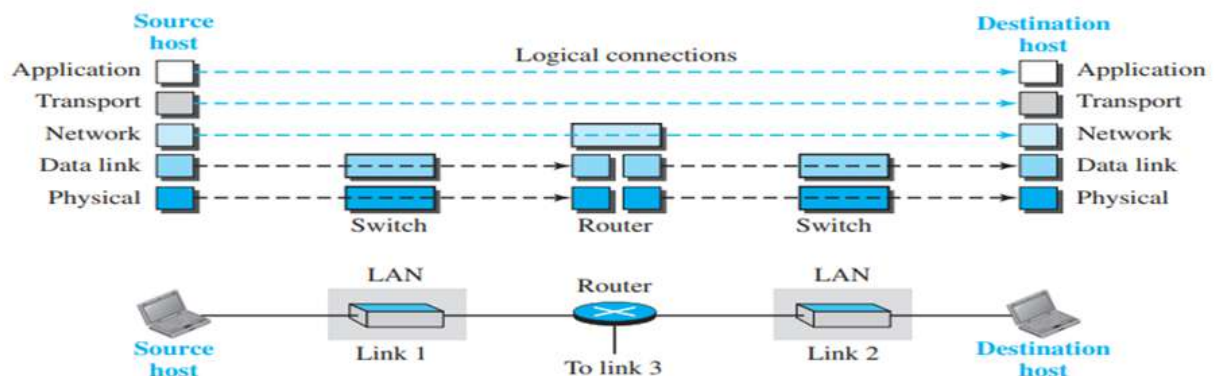


Figure 6 Logical connection between layers of TCP/IP Model

Encapsulation and De-encapsulation

One of the important concepts in protocol layering on the Internet is encapsulation/ de-encapsulation.

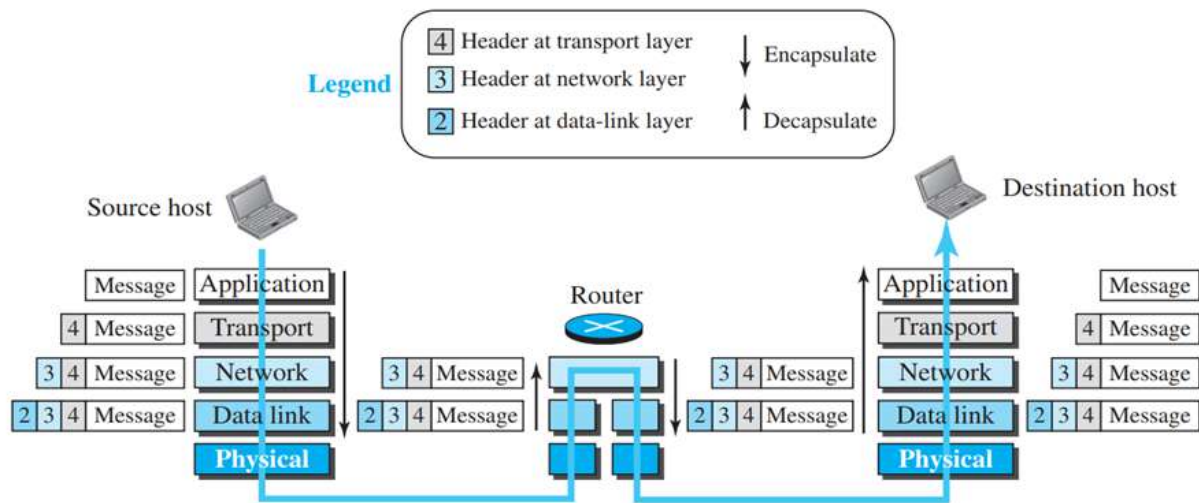


Figure 7 Encapsulation and De-encapsulation in TCP/IP Model

OSI versus TCP/IP

When we compare the two models, we find that two layers, session and presentation, are missing from the TCP/IP protocol suite. These two layers were not added to the TCP/IP protocol suite after the publication of the OSI model. The application layer in the suite is usually considered to be the combination of three layers in the OSI model, as shown below

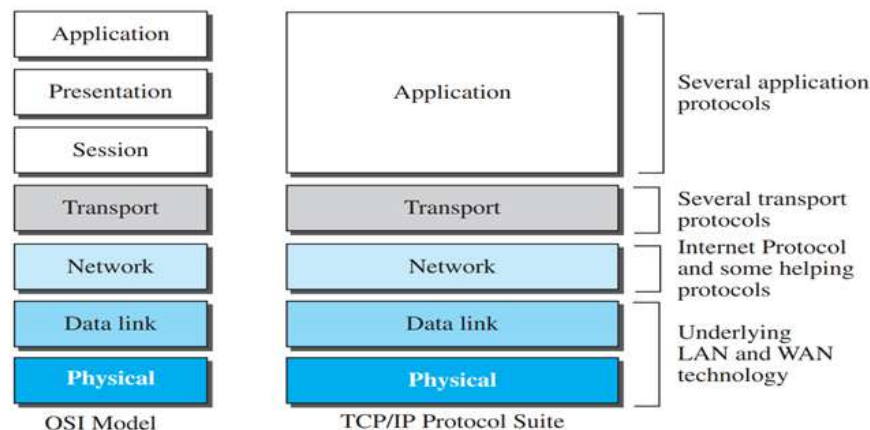


Figure 8. OSI Model vs TCP/IP

Two reasons were mentioned for this decision. First, TCP/IP has more than one transport-layer protocol. Some of the functionalities of the session layer are available in some of the transport-layer protocols. Second, the application layer is not only one piece of software. Many applications can be developed at this layer. If some of the functionalities mentioned in the session and presentation layers are needed for a particular application, they can be included in the development of that piece of software.

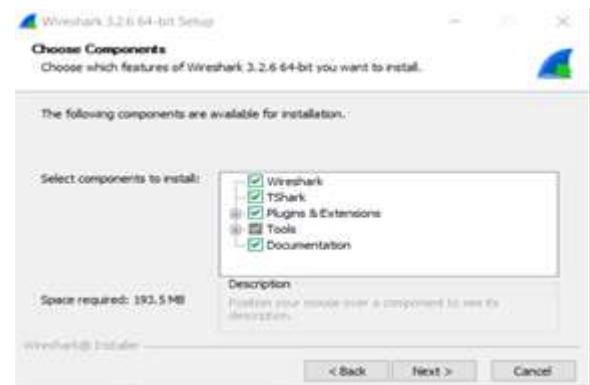
Network Simulator Setup

Before installing network simulator (eNSP) we need to **install the 3 prerequisites** required for eNSP

1. Installing WinPcap Driver



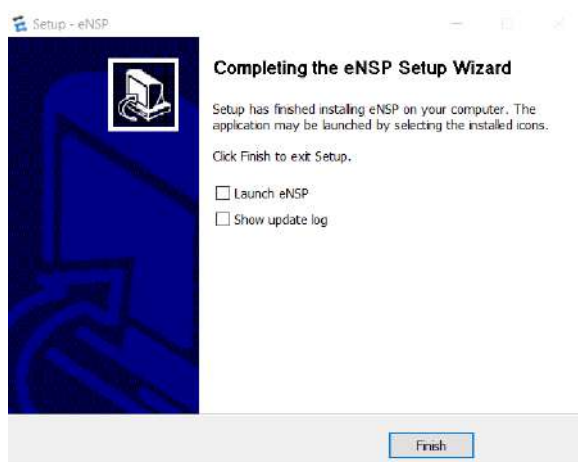
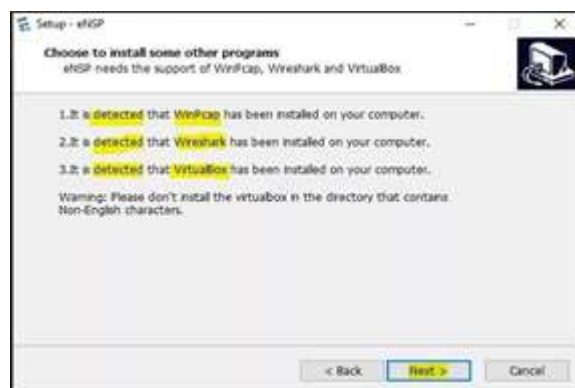
2. Installing Wireshark



3. Installing Oracle VirtualBox



4. Installing eNSP



Lab # 02

Objective

Understand and calculate the IP Addresses and its subnetting.

Theory

Internet Protocol (IP)

The network layer in version 4 can be thought of as one main protocol and three auxiliary ones. The main protocol, Internet Protocol version 4 (IPv4), is responsible for packetizing, forwarding, and delivery of a packet at the network layer. The Internet Control Message Protocol version 4 (ICMPv4) helps IPv4 to handle some errors that may occur in the network-layer delivery. The Internet Group Management Protocol (IGMP) is used to help IPv4 in multicasting. The Address Resolution Protocol (ARP) is used to glue the network and data-link layers in mapping network-layer addresses to link-layer addresses

IPv4

IPv4 is an IP version widely used to identify devices on a network using an addressing system. It was the first version of IP deployed for production in the ARPANET in 1983. It uses a 32-bit address scheme to store 2^{32} addresses which is more than 4 billion addresses. It is considered the primary Internet Protocol and carries 94% of Internet traffic.

Following are the features of IPv4:

- Connectionless Protocol
- Allow creating a simple virtual communication layer over diversified devices
- It requires less memory, and ease of remembering addresses
- Already supported protocol by millions of devices
- Offers video libraries and conferences

IPv4 Address Example				
Decimal Notation	131 . 153 . 40 . 106			
Binary Notation	10000011	10011001	00101000	1101010
	1 byte = 8 bits (Octet)	1 byte = 8 bits (Octet)	1 byte = 8 bits (Octet)	1 byte = 8 bits (Octet)
32 bits (4x8) = 4 bytes				

Figure 1. IPV4

IPv6

IPv6 is the most recent version of the Internet Protocol. This new IP address version is being deployed to fulfill the need for more Internet addresses. It was aimed to resolve issues that are associated with IPv4. With 128-bit address space, it allows 340 undecillion unique address space. IPv6 is also called IPng (Internet Protocol next generation).

Here are the features of IPv6:

- Hierarchical addressing and routing infrastructure
- Stateful and Stateless configuration
- Support for quality of service (QoS)
- An ideal protocol for neighboring node interaction

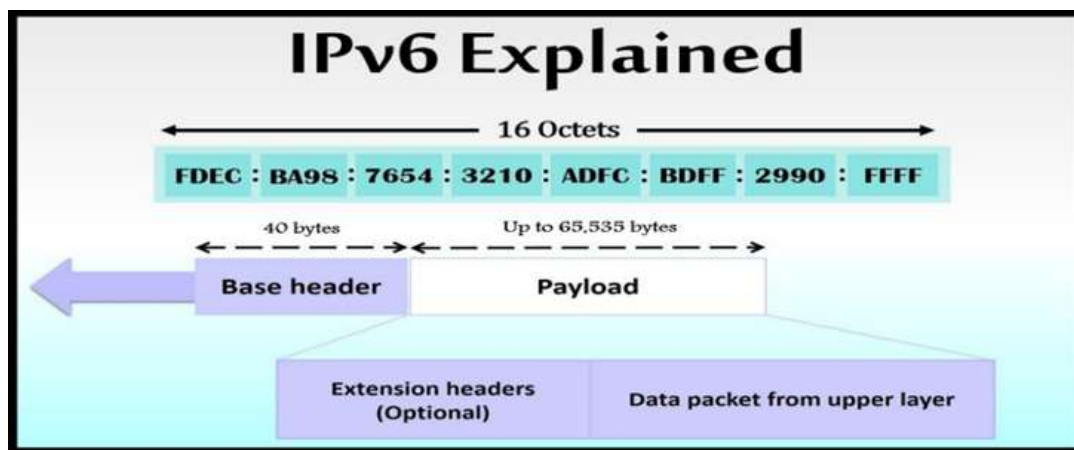


Figure 2. IPV6

Key Differences Between IPv4 and IPv6

IPv4 & IPv6 are both IP addresses that are binary numbers. IPv4 is a 32-bit binary number, and IPv6 is a 128-bit binary number address. IPv4 addresses are separated by periods, while IPv6 addresses are separated by colons. Both IP addresses are used to identify machines connected to a network. In principle, they are almost similar, but they are different in how they work.

Private and Public IP Addressing

The main difference between public and private IP addresses is how far they reach, and what they're connected to.

A **public IP address** identifies you to the wider internet so that all the information you're searching for can find you.

A **private IP address** is used within a private network to connect securely to other devices within that same network.

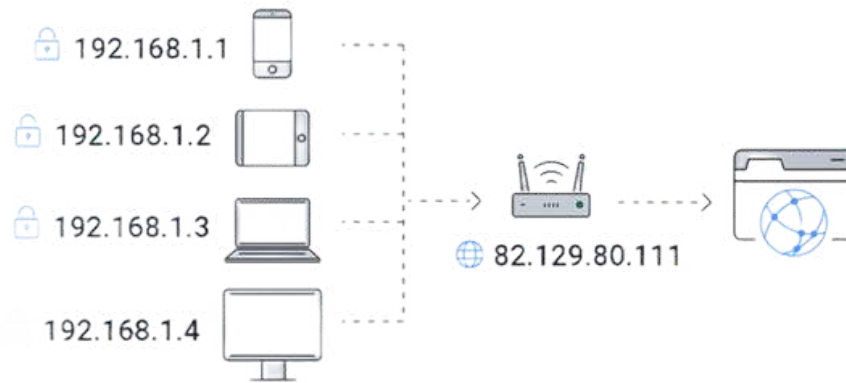


Figure 3. Public and Private IP Addressing

Private and Public IP Address Ranges

Your private IP address exists within specific private IP address ranges reserved by the Internet Assigned Numbers Authority (IANA) and should never appear on the internet. There are millions of private networks across the globe, all of which include devices assigned **private IP addresses** and the public IP address range encompasses every number *not* reserved for the private IP range.

Private	Public
Class A: 10.0.0.0 – 10.255.255.255	Class A: 1.0.0.0 – 9.255.255.255 11.0.0.0 – 126.255.255.255
Class B: 172.16.0.0 – 172.31.255.255	Class B: 128.0.0.0 – 172.15.255.255 172.32.0.0 – 191.255.255.255
Class C: 192.168.0.0 – 192.168.255.255	Class C: 192.0.0.0 – 192.167.255.255 192.169.0.0 – 223.255.255.255

Subnet Mask

A subnet mask is a 32-bit number created by setting host bits to all 0s and setting network bits to all 1s. In this way, the subnet mask separates the IP address into the network and host addresses.

The “255” address is always assigned to a broadcast address, and the “0” address is always assigned to a network address. Neither can be assigned to hosts, as they are reserved for these special purposes.

The IP address, subnet mask and gateway or router comprise an underlying structure—the Internet Protocol—that most networks use to facilitate inter-device communication.

When organizations need additional sub networking, subnetting divides the host element of the IP address further into a subnet. The goal of subnet masks is simply to enable the subnetting process. The phrase “mask” is applied because the subnet mask essentially uses its own 32-bit number to mask the IP address.

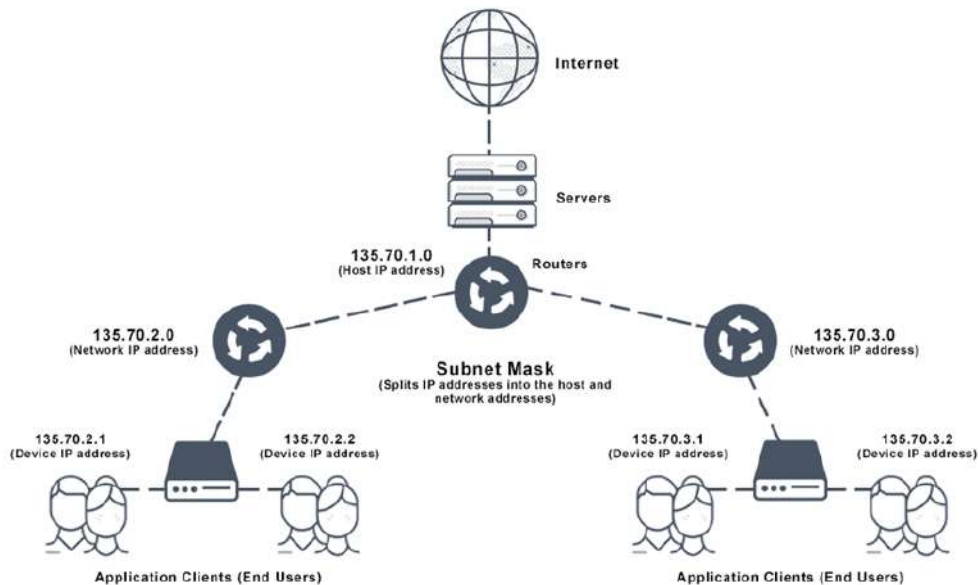


Figure 4. Subnet Mask

VLSM

VLSM stands for Variable Length Subnet Mask where the subnet design uses more than one mask in the same network which means more than one mask is used for different subnets of a single class A, B, C or a network. It is used to increase the usability of subnets as they can be of variable size. It is also defined as the process of subnetting of a subnet.

Earlier, it was required to use the same subnet mask across the network. This was called classful networking. With increase in complexity of networks and decrease in available IP addresses it became obvious that classful networking causes waste of valuable IP addresses. To understand how, consider Figure 1. The largest subnet requires 30 host addresses. So, across the network a mask of /27 is used, which gives 30 hosts per subnet. You will notice that in every subnet except the subnet attached to RouterD, some host addresses will remain unused. In particular, 28 host addresses are wasted for each link between the routers. In total this network wastes 118 addresses and uses 92 addresses.

To avoid wasting of IP addresses, classless networking was introduced by way of VLSM. VLSM allows you to use different subnet masks across the network for the same class of addresses. For example, a /30 subnet mask, which gives 2 host addresses per subnet, can be used for point-to-point links between routers. Figure 6 shows how VLSM can be used to save address space in the network shown in Figure 5.

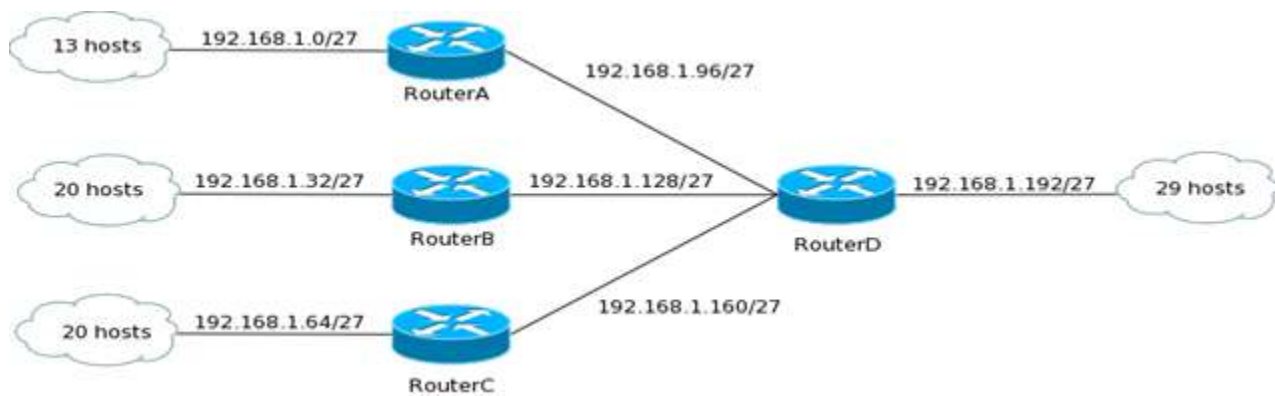


Figure 5. Without VLSM

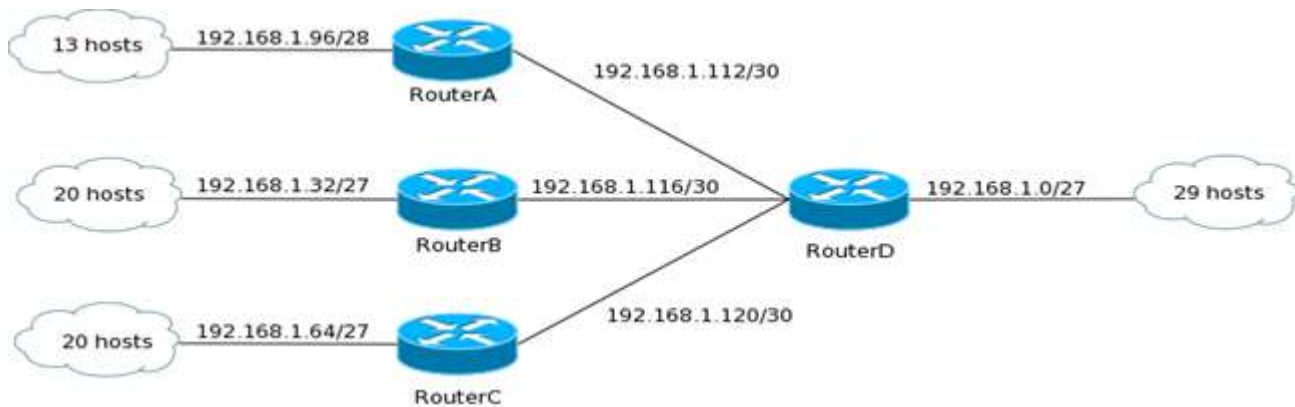


Figure 6. With VLSM

Subnetting and Supernetting

- **Subnetting**

Subnetting is the procedure to divide the network into sub-networks or small networks.

- **Supernetting**

Supernetting is the procedure to combine the small networks into larger space.

In subnetting, Network address bits are increased. On the other hand, in supernetting, Host address bits are increased. Subnetting is implemented via Variable-length subnet masking, while supernetting is implemented via Classless interdomain routing.

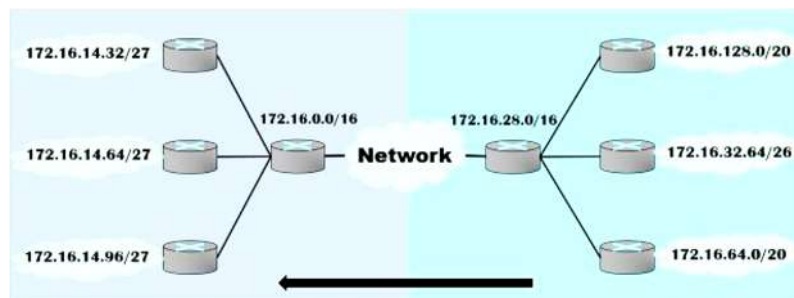


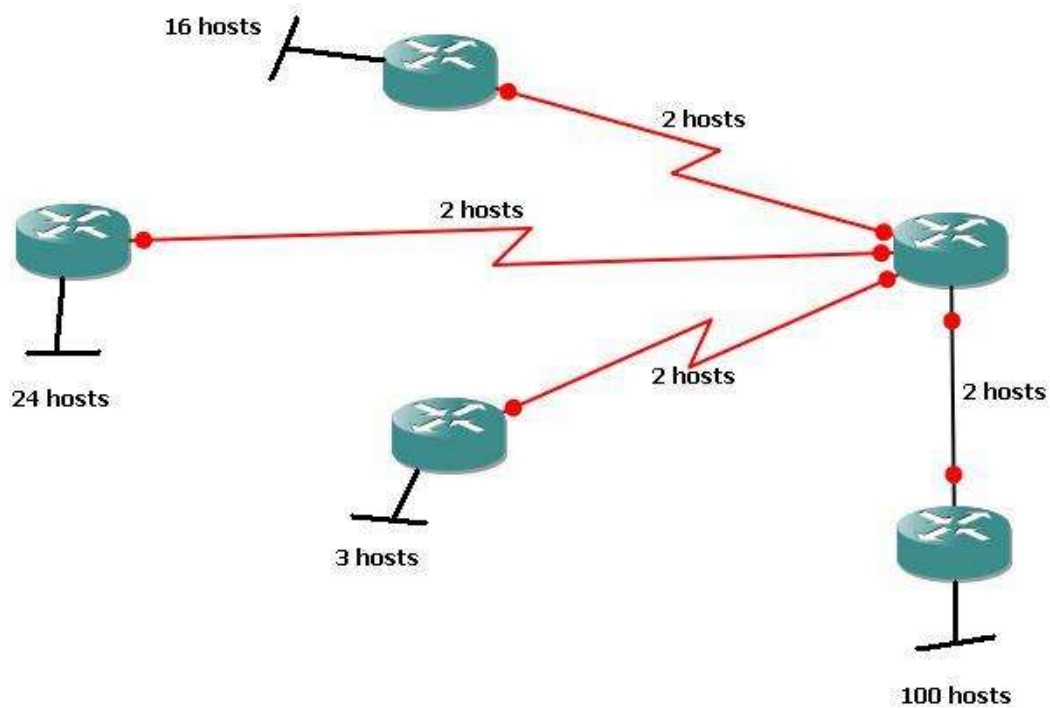
Figure 7. Subnetting vs Supernetting

Differences between Subnetting and Supernetting

	Subnetting	Supernetting
1.	Subnetting is the procedure to divide the network into sub networks.	While supernetting is the procedure of combining the small networks.
2.	In subnetting, Network address bits are increased.	While in supernetting, Host address bits are increased.
3.	In subnetting, the mask bits are moved towards the right.	While in supernetting, the mask bits are moved towards the left.
4.	Subnetting is implemented via Variable-length subnet masking.	While supernetting is implemented via Classless interdomain routing.
5.	In subnetting, Address depletion is either reduced or removed.	It is used to simplify the routing process.

Lab Task (Lab # 02):

Calculate the VLSM (IPs) using the Available Subnet – 24.23.5.0/24. Then implement in the following scenario with eNSP.



Lab # 03

Objective

Mac Addresses, ARP and Switching.

Theory

Address Resolution Protocol (ARP)

The network Address Resolution Protocol (ARP) is a procedure for mapping a dynamic IP address to a permanent physical machine address in a local area network (LAN). The physical machine address is also known as a media access control (MAC) address.

The job of ARP is essentially to translate 32-bit addresses to 48-bit addresses and vice versa. This is necessary because IP addresses in IP version 4 (IPv4) are 32 bits, but MAC addresses are 48 bits.

ARP works between Layers 2 and 3 of the Open Systems Interconnection model (OSI model). The MAC address exists on Layer 2 of the OSI model, the data link layer. The IP address exists on Layer 3, the network layer.

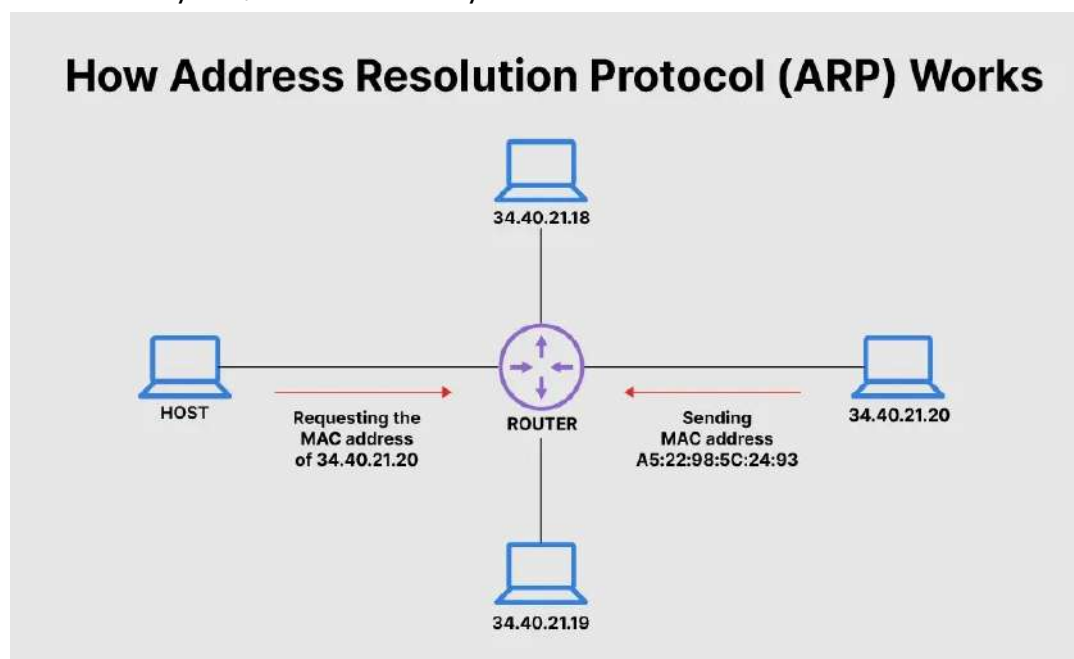


Figure 1. Working of ARP

There are four types of arp messages that may be sent by the arp protocol. These are identified by four values in the "operation" field of an arp message. The types of message are:

1. ARP-Request (Broadcast, source IP address of the requester)
2. ARP-Reply (Unicast to requester, the target)

Types of ARP

There are different versions and use cases of ARP. Let us take a look at a few.

Proxy ARP

Proxy ARP is a technique by which a proxy device on a given network answers the ARP request for an IP address that is not on that network. The proxy is aware of the location of the traffic's destination and offers its own MAC address as the destination.

Gratuitous ARP

Gratuitous ARP is almost like an administrative procedure, carried out as a way for a host on a network to simply announce or update its IP-to-MAC address. Gratuitous ARP is not prompted by an ARP request to translate an IP address to a MAC address.

Reverse ARP (RARP)

Host machines that do not know their own IP address can use the Reverse Address Resolution Protocol (RARP) for discovery.

Inverse ARP (IARP)

Whereas ARP uses an IP address to find a MAC address, IARP uses a MAC address to find an IP address.

Media Access Control Address (MAC Address)

A MAC address is a hardware identification number that uniquely identifies each device on a network. The MAC address is manufactured into every network card, such as an Ethernet card or Wi-Fi card, and therefore cannot be changed.

Because there are millions of networkable devices in existence, and each device needs to have a unique MAC address, there must be a very wide range of possible addresses. For this reason, MAC addresses are made up of six two-digit hexadecimal numbers, separated by colons. For example, an Ethernet card may have a MAC address of 00:0d:83:b1:c0:8e. Fortunately, you do not need to know this address, since it is automatically recognized by most networks.

Reason to have both IP and MAC addresses.

Both MAC and IP addresses are operated on different layers of the internet protocol suite. The MAC address works on layer 2 and helps identify the devices within the same broadcast network (such as the router). On the other hand, the IP addresses are used on layer 3 and help identify the devices on different networks.

MAC vs IP Address

Both the MAC address and IP address are the way to identify the device on the network.

IP Address	MAC Address
stands for Internet Protocol.	stands for Media Access Control.
It is the logical address provided by the ISP or Internet Service Provider.	It is the unique address provided by the manufacturer.
It is the logical address that identifies a network or device on the internet.	It is the physical address of the device's NIC that is used to identify a device within a network.
It operates on a network Layer.	It operates on the data link layer.
It is of 4 bytes for IPv4 and 8 bytes for IPv6 addresses.	It is the 6 -bytes hexadecimal address.

MAC Address Format

We cannot assign the MAC address to the device's NIC; it is preconfigured by the manufacturers.

- It is 12 digits or 6-byte hexadecimal number, which is represented in colon-hexadecimal notation format. It is divided into six octets, and each octet contains 8 bits.
- The first three octets are used as the OUI or Organisationally Unique Identifier. These MAC prefixes are assigned to each organization or vendor by the IEEE Registration Authority Committee.

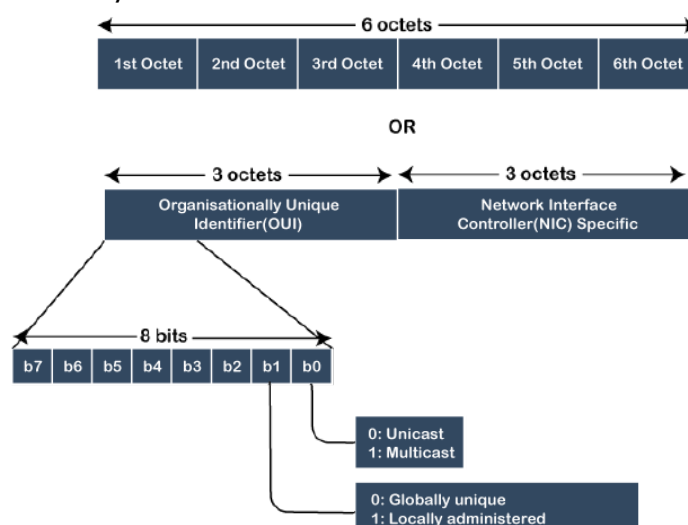


Figure 2. Mac Address Format

Why should the MAC address be unique in the LAN network?

If a LAN network has two or more devices with the same MAC address, that network will not work.

Suppose three devices A, B, and C are connected to a network through a switch. The MAC addresses of these devices are 11000ABB28FC, 00000ABB28FC, and 00000ABB28FC, respectively. The NIC of devices B and C have the same MAC address. If device A sends a data frame to the address 00000ABB28FC, the switch will fail to deliver this frame to the destination, as it has two recipients of this data frame.

What is a network switch?

A network switch connects devices within a network (often a local area network, or LAN*) and forwards data packets to and from those devices. Unlike a router, a switch only sends data to the single device it is intended for (which may be another switch, a router, or a user's computer), not to networks of multiple devices.

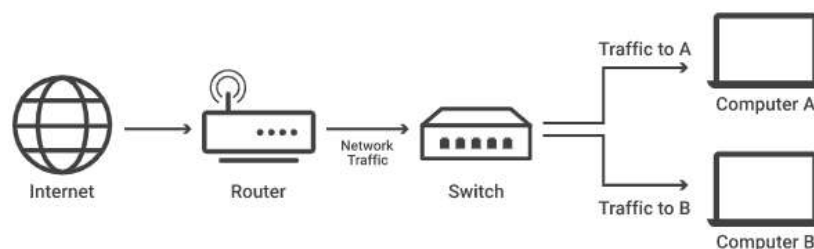


Fig 1. A local area network (LAN) is a group of connected devices within close physical proximity.

What is the difference between a switch and a router?

Routers select paths for data packets to cross networks and reach their destinations. In practice, what this means is that routers are necessary for an Internet connection, while switches are only used for interconnecting devices. Homes and small offices need routers for Internet access, but most do not need a network switch, unless they require a large amount of Ethernet* ports. However, large offices, networks, and data centers with dozens or hundreds of computers usually do require switches.

What is a layer 2 switch and layer 3 switch?

Network switches can operate at either OSI layer 2 (the data link layer) or layer 3 (the network layer). Layer 2 switches forward data based on the destination MAC address (see below for definition), while layer 3 switches forward data based on the destination IP address. Some switches can do both.

Most switches, however, are layer 2 switches. Layer 2 switches most often connect to the devices in their networks using Ethernet cables. Ethernet cables are physical cables that plug into devices via Ethernet ports.

What is an unmanaged switch? What is a managed switch?

An unmanaged switch simply creates more Ethernet ports on a LAN, so that more local devices can access the Internet. Unmanaged switches pass data back and forth based on device MAC addresses.

A managed switch fulfills the same function for much larger networks, and offers network administrators much more control over how traffic is prioritized. They also enable administrators to set up Virtual LANs (VLANs) to further subdivide a local network into smaller chunks.

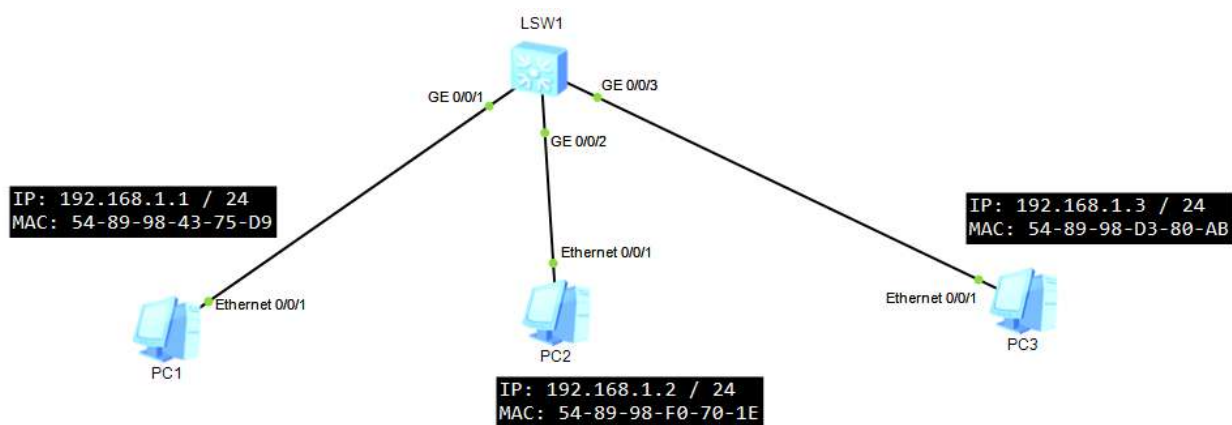
How do network switches know the MAC addresses of the devices in their network?

In layer 2, Network Switches maintains a MAC address table known as Content Addressable Memory (CAM) table that contains MAC Address and port number. Switches follow this simple algorithm:

1. When a frame is received, the switch compares the Source MAC address to the MAC address table. If the Source is unknown, the switch adds it to the table along with the port number the packet was received on. In this way, the switch learns the MAC address and port of every transmitting device.
2. The switch then compares the Destination MAC address with the table. If there is an entry, the switch forwards the frame out the associated port. If there is no entry, the switch sends the packet out all its ports, except the port that the frame was received on (Flooding).

Lab Task (Lab # 03):

Create the following scenario in eNSP with the mentioned IP addresses. Please note the MAC address may vary system to system, mention your MAC addresses. Ping PC3 from PC1 and explain how the ARP table would be maintained and what would be the role of PC2 in this process. Also explain the difference between the static and dynamic ARP entries.



Lab # 04

Objective

Understand Virtual Local Area Network (VLAN).

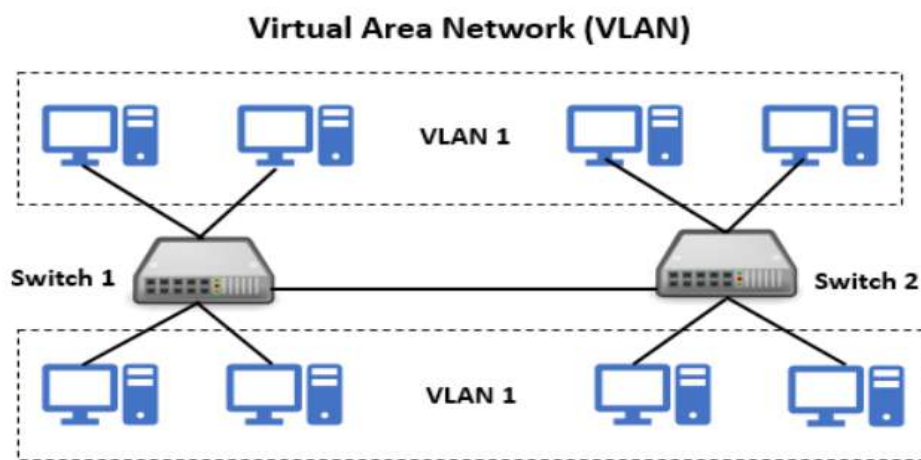
Theory

VLAN

VLANs (Virtual LANs) are logical grouping of devices in the same broadcast domain. VLANs are usually configured on switches by placing some interfaces into one broadcast domain and some interfaces into another. Each VLAN acts as a subgroup of the switch ports in an Ethernet LAN.

VLANs can spread across multiple switches, with each VLAN being treated as its own subnet or broadcast domain. This means that frames broadcasted onto the network will be switched only between the ports within the same VLAN.

A VLAN acts like a physical LAN, but it allows hosts to be grouped together in the same broadcast domain even if they are not connected to the same switch.



Function of VLAN

The function is to separate layer 2 traffic. When the host in one VLAN is unable to communicate data between the host in another VLAN, a router is placed to pass the data between these two. We can connect all the hosts in one switch to connect each group, but the cost is prohibitive, so it is preferred for VLANs to broadcast the data. Also, the other best reason is for the secure data transfer between two hosts. The only members in the VLANs can access the network and data.

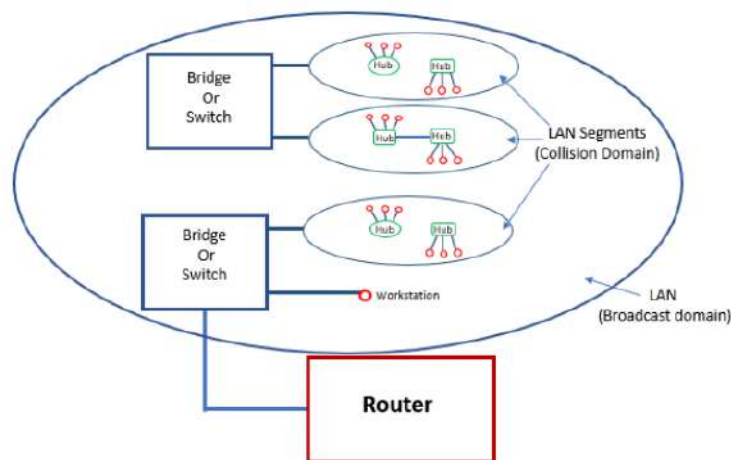


Figure 2. Function of VLAN

How VLAN works

Here is step by step details of how VLAN works:

- VLANs in networking are identified by a number.
- A Valid range is 1-4094. On a VLAN switch, you assign ports with the proper VLAN number.
- The switch then allows data which needs to be sent between various ports having the same VLAN.
- Since almost all networks are larger than a single switch, there should be a way to send traffic between two switches.
- One simple and easy way to do this is to assign a port on each network switch with a VLAN and run a cable between them.

VLAN Tag

Vlan Tags help identify the VLAN to which a received frame belongs. IEEE 802.1Q defines a 4-byte VLAN tag for Ethernet frames, enabling switches to identify the VLANs to which received frames belong.

Destination MAC address	Source MAC address	Length/ Type	Data	FCS
-------------------------	--------------------	-----------------	------	-----

Figure 3. Untagged Frame

Destination MAC address	Source MAC address	Tag	Length/ Type	Data	FCS
-------------------------	--------------------	-----	-----------------	------	-----

Figure 4. Tagged Frame

Components of VLAN Tag

- Tag protocol identifier (TPID): identifies the type of a frame. The value 0x8100 indicates an IEEE 802.1Q frame.

- PRI: identifies the priority of a frame, which is mainly used for QoS.
- Canonical format indicator (CFI): indicates whether a MAC address is in the canonical format. For Ethernet frames, the value of this field is 0.
- VLAN ID: identifies the VLAN to which a frame belongs.



Figure 5. Vlan Tag Components

Types of VLANs

Here are the important types of VLANs

- Port-Based VLAN
- Protocol-Based VLAN
- MAC-Based VLAN

Port-Based VLAN

Port-based VLANs group virtual local area networks by port. In this type of virtual LAN, a switch port can be configured manually to a member of VLAN.

Devices that are connected to this port will belong to the same broadcast domain that is because all other ports are configured with a similar VLAN number.

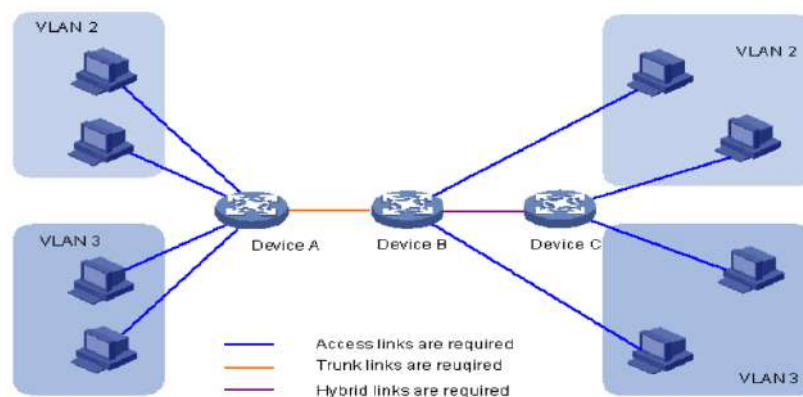


Figure 6. Port-Based VLAN

Protocol Based VLAN

This type of VLAN processes traffic based on a protocol that can be used to define filtering criteria for tags, which are untagged packets.

In this VLAN, the layer-3 protocol is carried by the frame to determine VLAN membership. It works in multi-protocol environments. This method is not practical in a predominately IP based network.

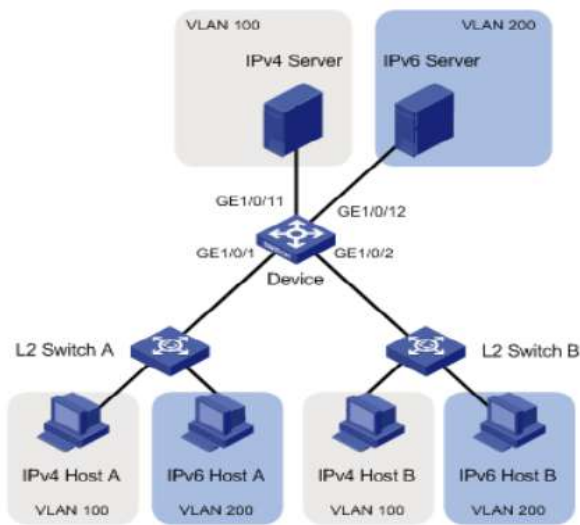


Figure 7. Protocol-Based VLAN

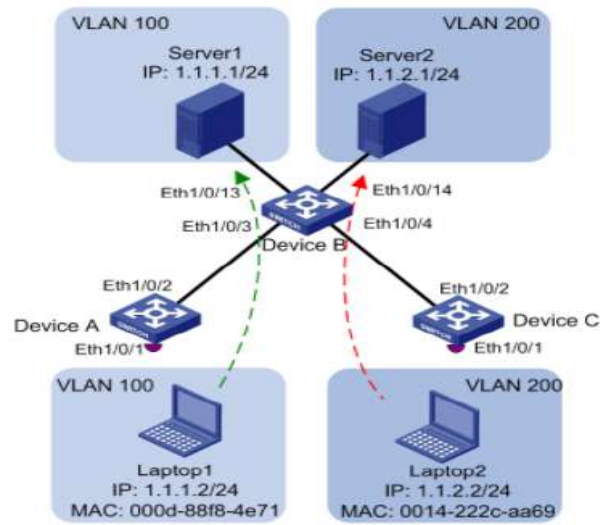


Figure 8. MAC-Based VLAN

MAC Based VLAN

MAC Based VLAN allows incoming untagged packets to be assigned virtual LAN and, thereby, classify traffic depending on the packet source address. You define a Mac address to VLAN mapping by configuring mapping the entry in MAC to the VLAN table.

This entry is specified using the source Mac address proper VLAN ID. The configurations of tables are shared among all device ports.

VLAN Ranges

Here are the important ranges of VLAN:

Range	Description
VLAN 0-4095	Reserved VLAN, which cannot be seen or used.
VLAN 1:	This is a default VLAN of switches. You cannot delete or edit this VLAN, but it can be used.
VLAN 2-1001:	It is a normal VLAN range. You can create, edit, and delete it.
VLAN 1002-1005:	These ranges are CISCO defaults for token rings and FDDI. You cannot delete this VLAN.
VLAN 1006-4094:	It is an extended range of VLANs.

Example of VLAN

In the below example, there are 6 hosts on 6 switches having different VLANs. You need 6 ports to connect switches together. It means, if you have 24 various VLANs, you will have only 24 hosts on 45 port switches.

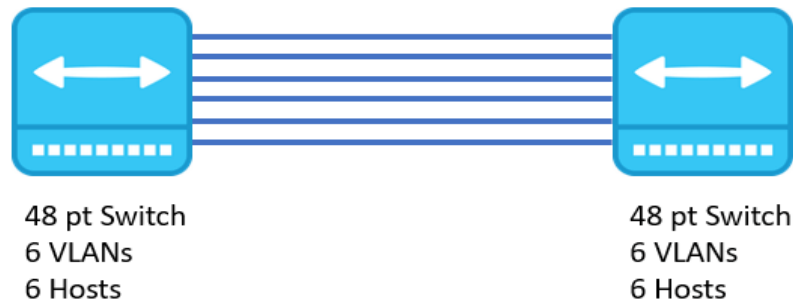


Figure 9. Example of VLAN

Application/Purpose of VLAN

Here are the important uses of VLAN:

- VLAN is used when you have 200+ devices on your LAN.
- It is helpful when you have a lot of traffic on a LAN.
- VLAN is ideal when a group of users need more security or are being slowed down by many broadcasts.
- It is used when users are not on one broadcast domain.
- Make a single switch into multiple switches.

Basic Configuration Commands

Command	Description
VLAN (To be used in system view)	
<code>vlan <i>vlan-id</i></code>	This command creates a VLAN and displays the VLAN view. If the VLAN to be created already exists, this command directly displays the VLAN view (value of <i>vlan-id</i> is an integer that ranges from 1 - 4096)
<code>vlan <i>batch</i></code> <code>{<i>vlan-id1</i> [to <i>vlan-id2</i>]}</code>	This command creates VLAN in a batch. In this command. • “batch” creates VLANs in a batch • “<i>vlan-id1</i>” specifies a start VLAN ID. • “<i>vlan-id2</i>” specifies an end VLAN ID.

Access Interface

(To be used in interface view)

port link-type access	In the interface view, set the link type of the interface to access.
port default vlan <i>vlan-id</i>	In the interface view, configure a default VLAN for the interface and add the interface to the VLAN. “vlan-id” specifies an ID for the default VLAN (value of vlan-id is an integer that ranges from 1 - 4096)

Trunk Interface

(To be used in interface view)

port link-type trunk	Set the link type of the interface to trunk.
port trunk allow-pass vlan { { <i>vlan-id1</i> [to <i>vlan-id2</i>] } all }	Add the trunk interface to specified VLANs.
port trunk pvid vlan vlan-id	In the interface view, configure a default VLAN for the trunk interface.

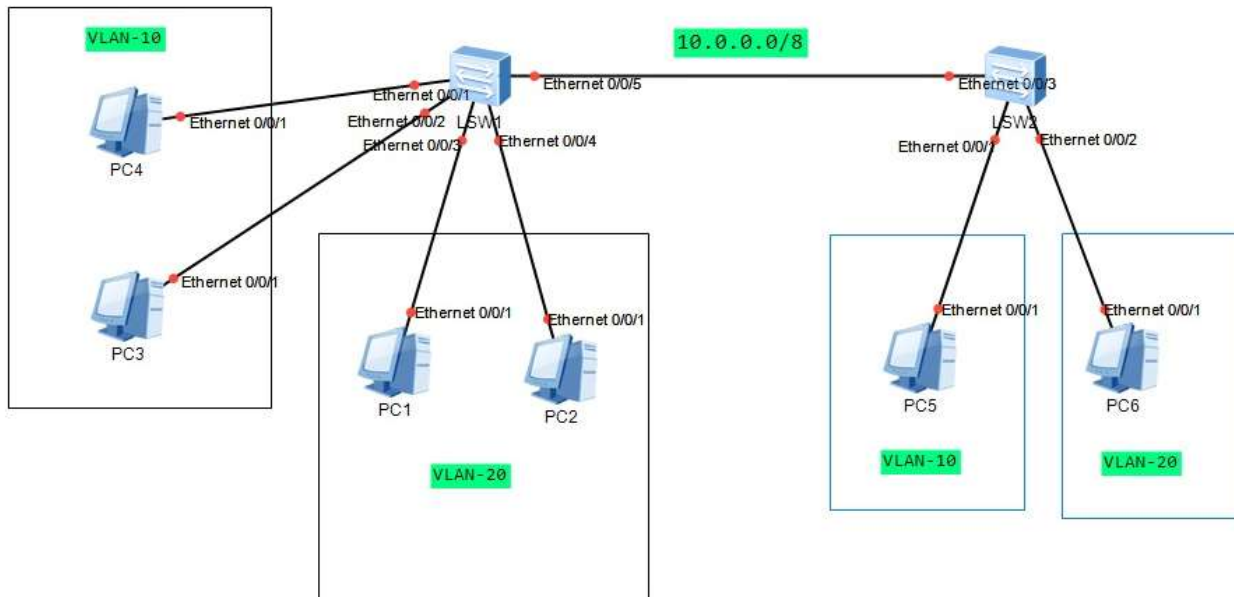
Hybrid Interface

(To be used in interface view)

port link-type hybrid	Set the link type of the interface to hybrid.
port hybrid untagged vlan { { <i>vlan-id1</i> [to <i>vlan-id2</i>] } all }	Add the hybrid interface to specified VLANs in untagged mode.
port hybrid tagged vlan { { <i>vlan-id1</i> [to <i>vlan-id2</i>] } all }	Add the hybrid interface to specified VLANs in tagged mode.
port hybrid pvid vlan vlan-id	Configure a default VLAN for the hybrid interface.

Lab Task (Lab # 04):

Implement the following VLAN scenario in eNSP simulator:



Lab # 05

Objective

VLAN Intercommunication Basics

Theory

Background

In real-world network deployments, different IP address segments are assigned to different VLANs.

PCs on the same network segment in the same VLAN can directly communicate with each other without the need for Layer 3 forwarding devices. This communication mode is called Layer 2 communication.

inter-VLAN communication belongs to Layer 3 communication, which requires Layer 3 devices. Common Layer 3 devices: routers, Layer 3 switches, firewalls, etc. Inter-VLAN communication is implemented by connecting a Layer 2 switch to a Layer 3 interface of a Layer 3 device. The communication packets are routed by the Layer 3 device.

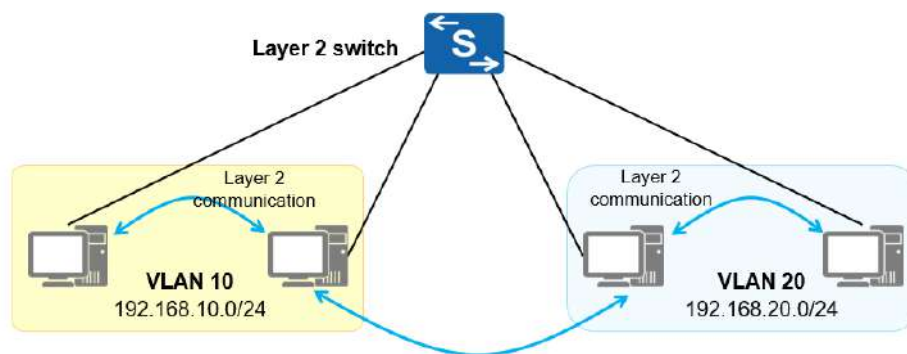


Figure 1. Layer-2 VLAN Communication

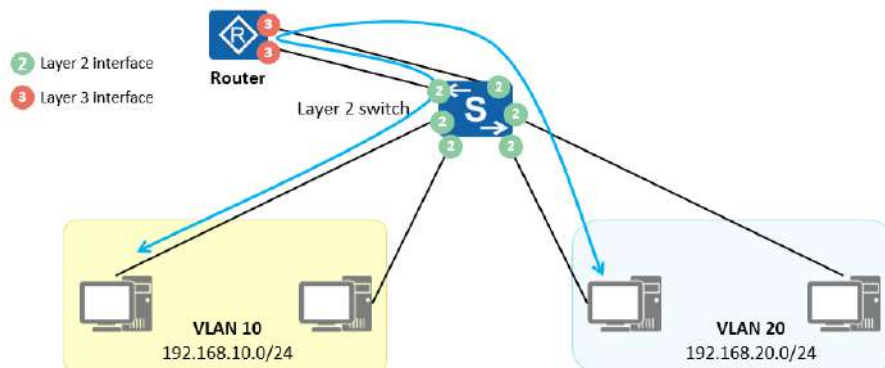


Figure 2. Layer-3 VLAN Communication

Using Router's Physical Interfaces

- The Layer 3 interfaces of the router function as gateways to forward traffic from the local network segment to other network segments.
- The Layer 3 interfaces of the router cannot process data frames with VLAN tags. Therefore, the interfaces of the switch connected to the router must be set to the access type.
- One physical interface of the router can function as the gateway of only one VLAN, meaning that the number of required physical interfaces are determined by the quantity of the deployed VLANs.
- A router, mainly forwarding packets at Layer 3, provides only a small number of physical interfaces. Therefore, the scalability of this solution is poor.

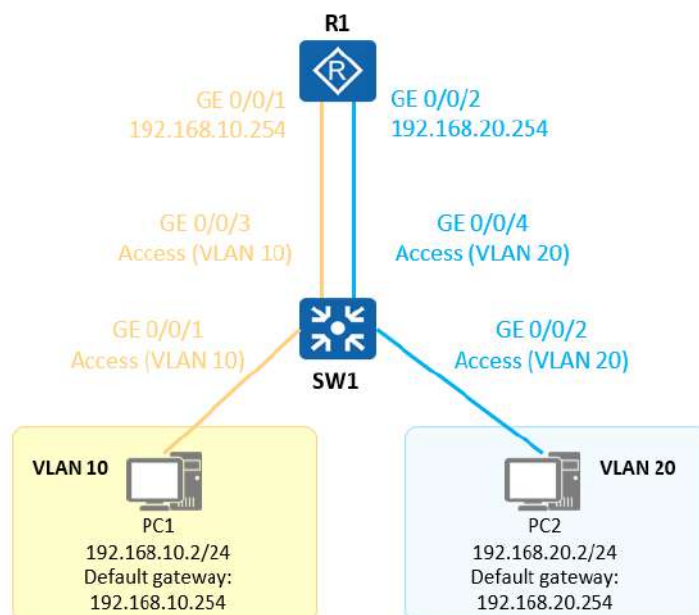
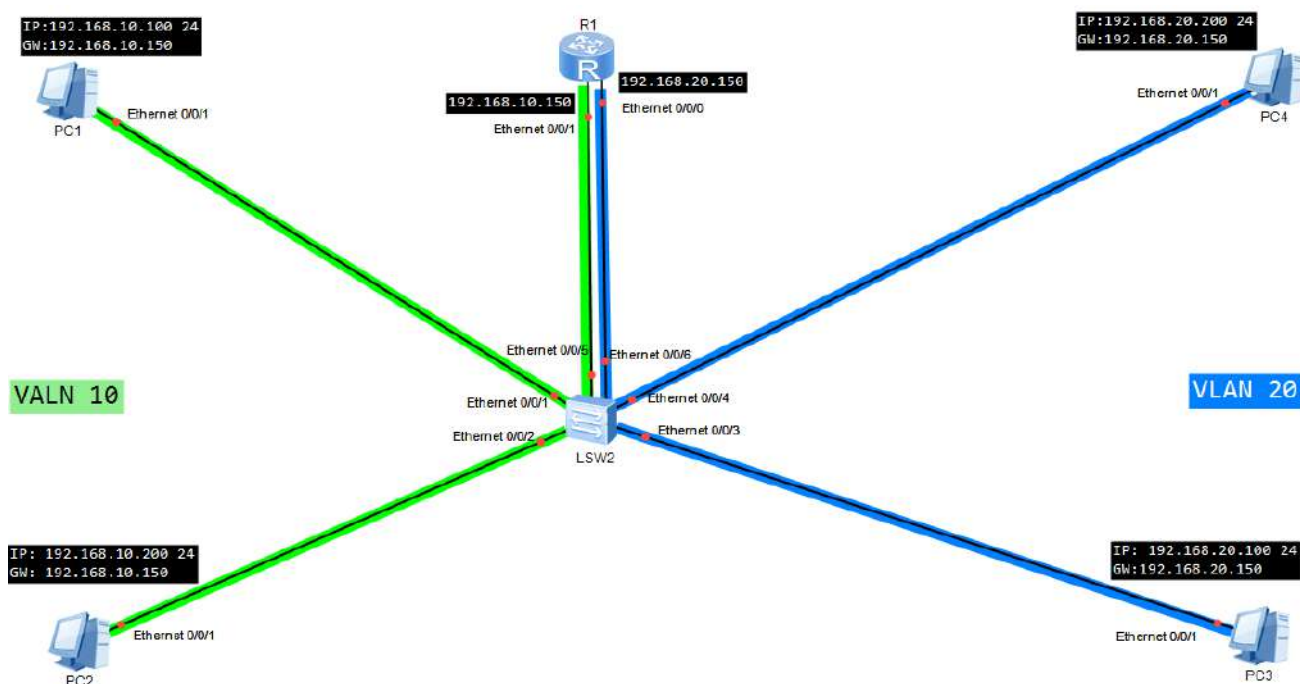


Figure 3. Using router's physical interfaces

Lab Task (Lab # 05):**Configure the following scenario in eNSP:**

Lab # 06

Objective

VLAN Intercommunication using Router's subinterface

Theory

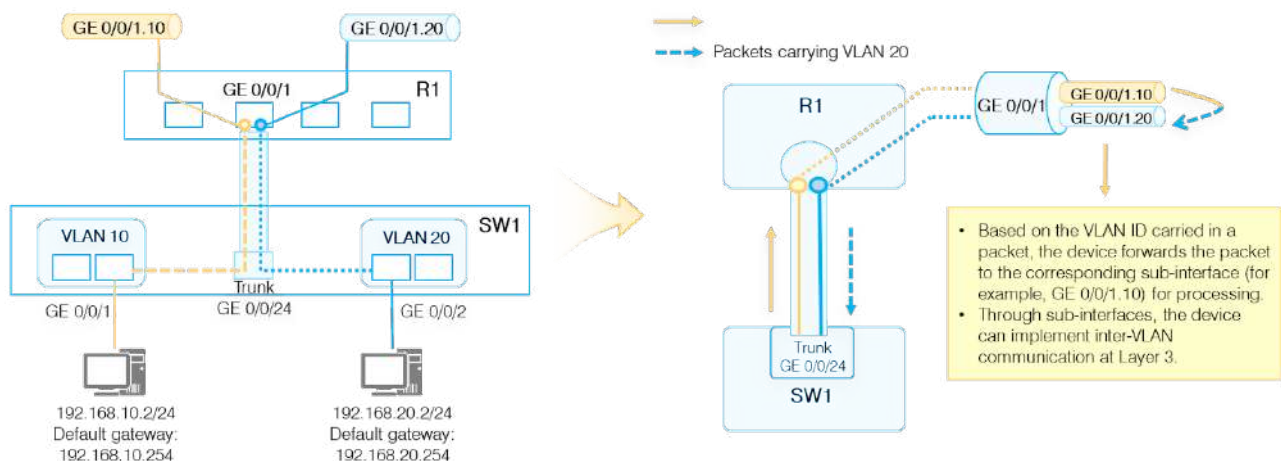
Router's Subinterfaces

A sub-interface is a logical interface created on a router's Ethernet interface and is identified by a physical interface number and a sub-interface number. Similar to a physical interface, a sub-interface can perform Layer 3 forwarding.

Different from a physical interface, a sub-interface can terminate data frames with VLAN tags. You can create multiple sub-interfaces on one physical interface. After connecting the physical interface to the trunk interface of the switch, the physical interface can provide Layer 3 forwarding services for multiple VLANs.

Subinterface Process

The interface connecting the switch to the router is set to a trunk interface. The router forwards the received packets to the corresponding sub-interfaces according to the VLAN tags in the packets.

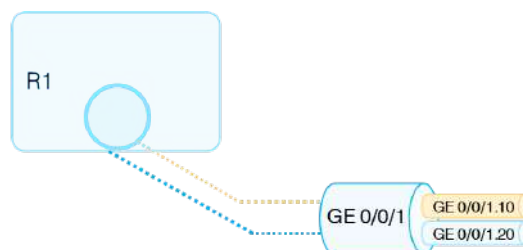


Commands for creating and configuring Sub-Interface

Command	Description
[R1]interface GigabitEthernet0/0/1.10	This command creates a sub-interface 1.10 on the physical interface GigabitEthernet0/0/1. For easy memorization, a sub-interface number is generally the same as the VLAN ID to be terminated on the sub-interface.

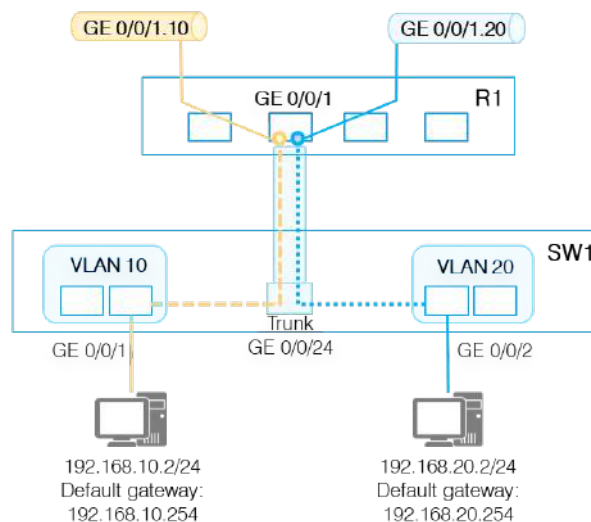
[R1-GigabitEthernet0/0/1.10] dot1q termination vid 10	Enable dot1q VLAN tag termination for single-tagged packets on the sub-interface.
[R1-GigabitEthernet0/0/1.10] ip address 192.168.10.254 24	Assigning an IP address to the sub-interface.
[R1-GigabitEthernet0/0/1.10] arp broadcast enable	By default ARP broadcast is not enabled on VLAN tag termination sub-interfaces. VLAN tag termination sub-interfaces cannot forward broadcast packets and automatically discard received ones.

The above steps can be repeated to create another sub-interface (GigabitEthernet0/0/1.20 in this case). The resulting structure would look like the image as follows:



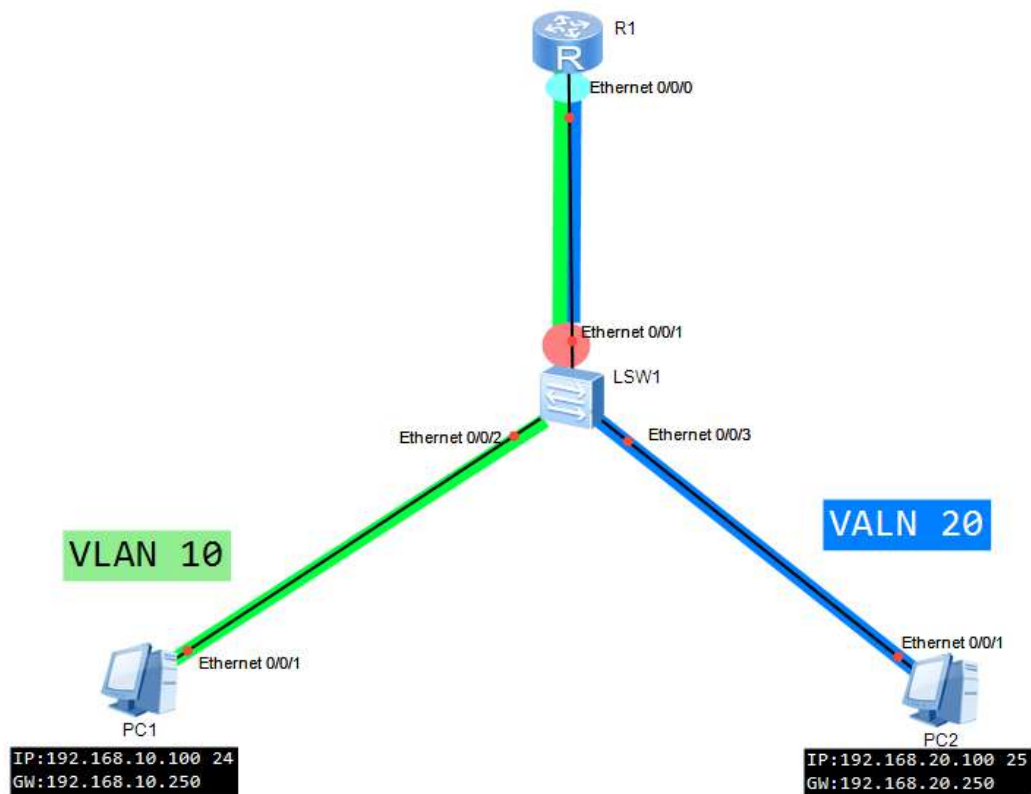
Router on a Stick

A router on a stick, also known as a one-armed router, is a router that has a single physical or logical connection to a network. It is a method of inter-VLAN (virtual local area networks) routing where one router is connected to a switch via a single cable. The interface connecting the switch to the router is set to a trunk interface. The router forwards the received packets to the corresponding sub-interfaces according to the VLAN tags in the packets.



Lab Task (Lab # 06):

Configure the following router-on-a-stick in eNSP.



Lab # 07

Objective

VLAN Intercommunication using VLANIF

Theory

Using Layer-3 switches

- A Layer 2 switch provides only Layer 2 switching functions.
- A Layer 3 switch provides routing functions through Layer 3 interfaces (such as VLANIF interfaces) as well as the functions of a Layer 2 switch.
- A VLANIF interface is a Layer 3 logical interface that can remove and add VLAN tags. VLANIF interfaces therefore can be used to implement inter-VLAN communication.
- A VLANIF interface number is the same as the ID of its corresponding VLAN. For example, VLANIF 10 is created based on VLAN 10.

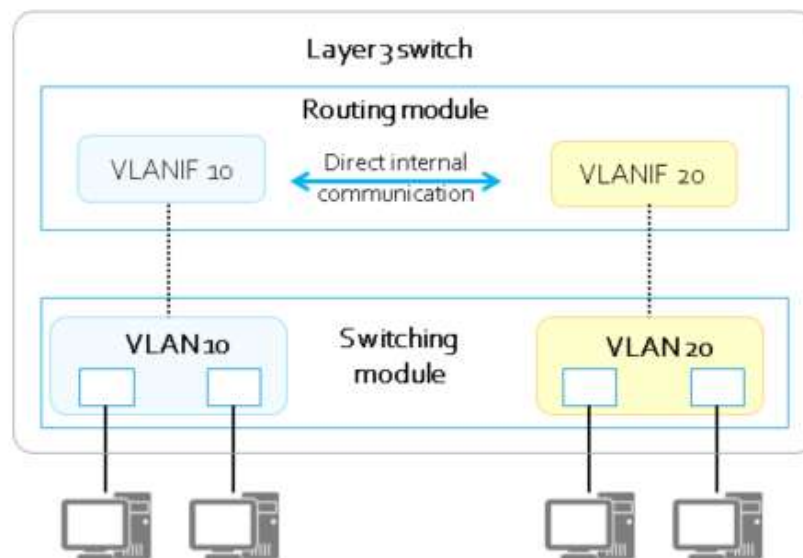


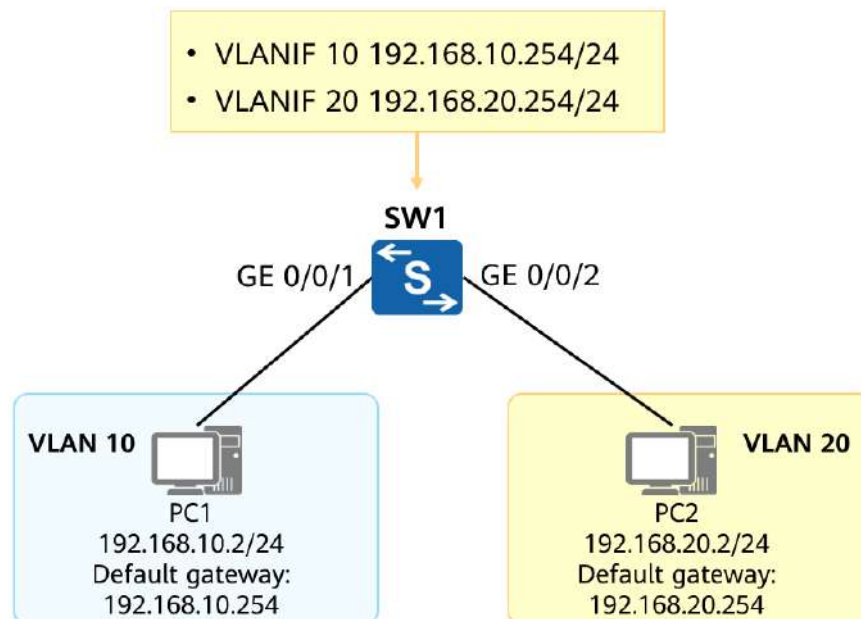
Figure 5. Layer-3 switch and VLANIF Interface

Example with configuration:Basic Configuration:

```
[SW1]vlan batch 10 20
[SW1] interface GigabitEthernet 0/0/1
[SW1-GigabitEthernet0/0/1] port link-type access
[SW1-GigabitEthernet0/0/1] port default vlan 10
[SW1] interface GigabitEthernet 0/0/2
[SW1-GigabitEthernet0/0/2] port link-type access
[SW1-GigabitEthernet0/0/2] port default vlan 20
```

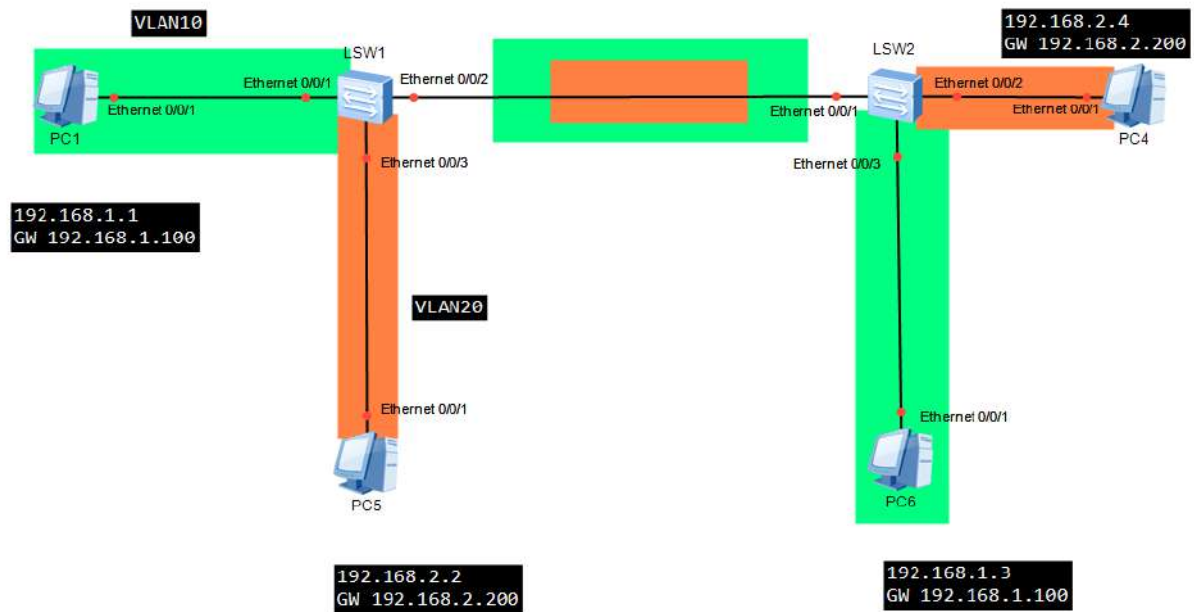
Vlanif Configuration:

```
[SW1]interface Vlanif 10
[SW1-Vlanif10]ip address 192.168.10.254 24
[SW1]interface Vlanif 20
[SW1-Vlanif20]ip address 192.168.20.254 24
```



Lab Task (Lab # 07):

Configure the following scenario (Inter VLAN routing) in eNSP.



Lab # 08

Objective

Static Routes

Theory

In computer networks, routing can be done with two methods. One of them is **Static Routing** and the other is **Dynamic Routing**. Static routing is the manual method of routing and Dynamic routing is the routing that is done with the help of Routing Protocols.

In Static routing, we can enter all the routes manually to the router. In other words, we can define each routing step one by one. To access a network, which nodes we need to pass through, we can define such steps. This work is not simple, so static routing is used in small networks. Beside, in large networks, static routing is also used beside dynamic routing. Think about it, defining all the routes in a large network, isn't it a very difficult job?

To configure a static route, we use three parameters. These parameters are given below:

- Destination Network IP Address
- Destination Subnet Mask
- Next Hop IP Address or Exit Interface

Destination Network IP Address shows the ip address of our destination network

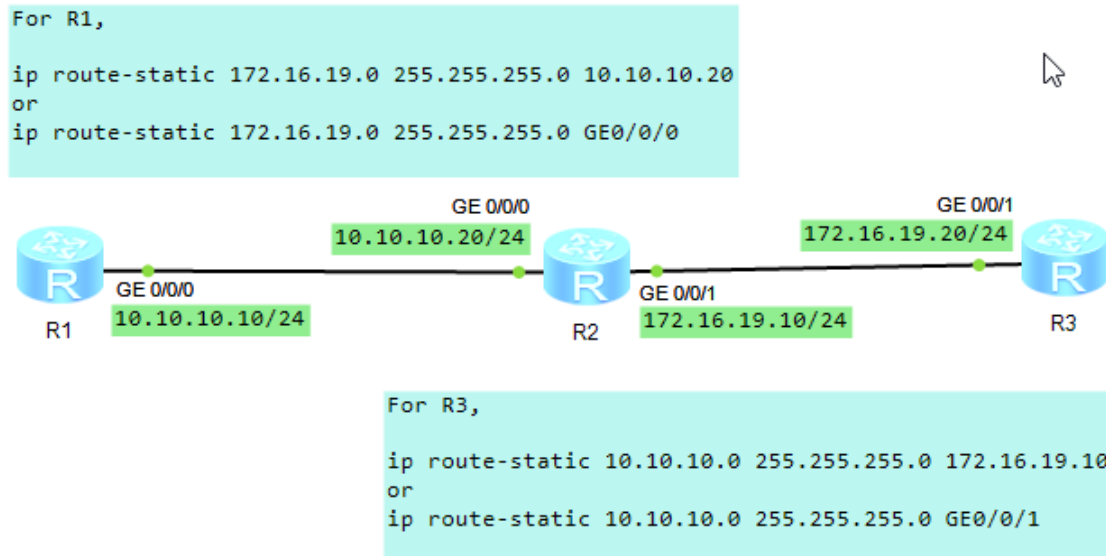
Destination Subnet Mask shows the destination network's subnet mask.

Next Hop IP Address or Exit Interface For the last parameter, we have two different ways. If we use *Next Hop IP address*, this shows the router that the traffic will go after our router. Here, we use the interface IP address of the router that we will pass through on the way to the destination.

If we use *Exit Interface*, we define the interface that we will use to go through the destination interface. This exit interface is one of the interfaces of our router.

Example

in the below topology, there are three routers and two networks. The end routers are directly connected to only one network, so they do not know the other network. So, with static routes, we need to teach the other network to each of them on both routers. So, we will configure static routes on Router A and RouterC.

**Figure 1. Static Routing**

Router A does not know the 172.16.19.0/24 network and Router C does not know 10.10.10.0/24 network. So, we will configure static routers through these networks.

As you can see above, to write a static route, first we will define the network that we would like to go and then we will write a subnet mask of this network. And lastly, we will define the next hop interface or the exit interface of the router.

So, why do we configure static routes in both directions? Because the end routers do not know the other network and the static routes are unidirectional routes. If we want communication between Router A and Router C, we should configure static routes on both routers.

Additional Codes:

[R1] display ip interface brief

[R1] display ip routing-table

Notes:

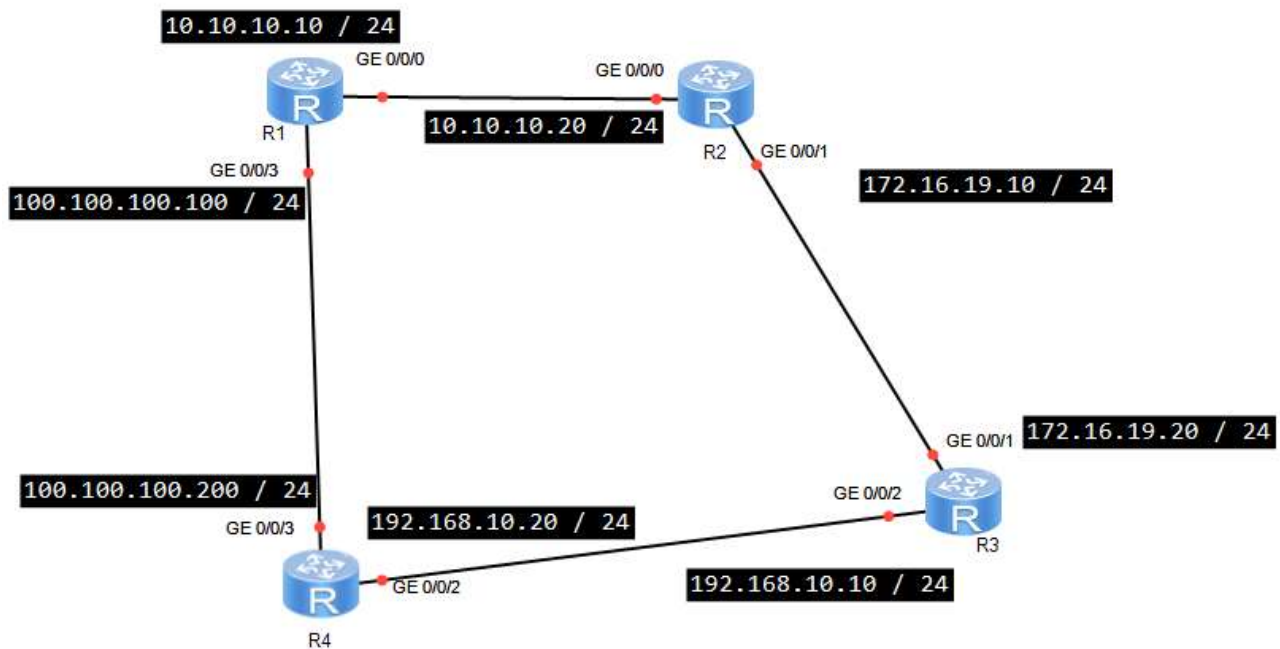
Default preference of static route is 60

Lower the preference, higher the chances of selecting a route.

Lab Task (Lab # 08):

Configure the following scenario by applying static routes in eNSP.

Note: All routers should be able to ping all interfaces.



Lab # 09

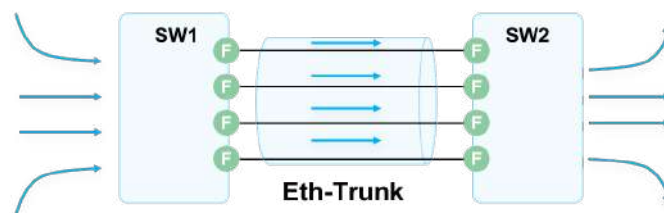
Objective

Overview of Link Aggregation (Eth-Trunk)

Theory

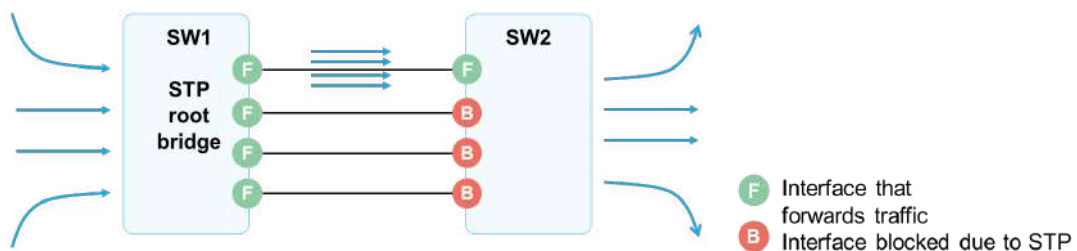
Definition

Ethernet link aggregation, also called Eth-Trunk, bundles multiple physical links to form a logical link to increase link bandwidth. The bundled links back up each other, increasing reliability.



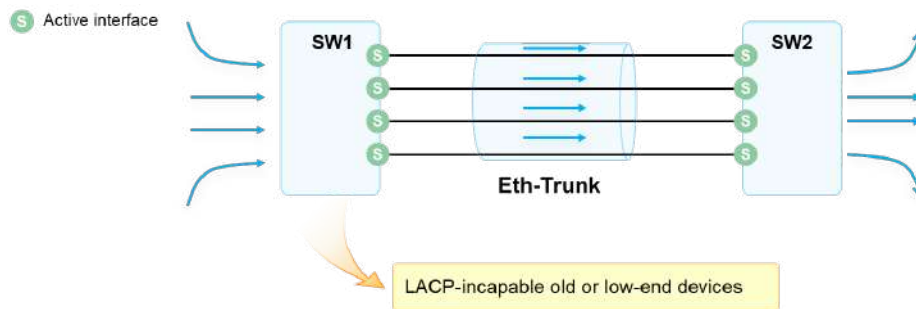
How Eth-Trunk improves bandwidth

- When multiple links exist between devices, traffic is forwarded on only one link due to STP. In this case, the inter-device link bandwidth remains unchanged.

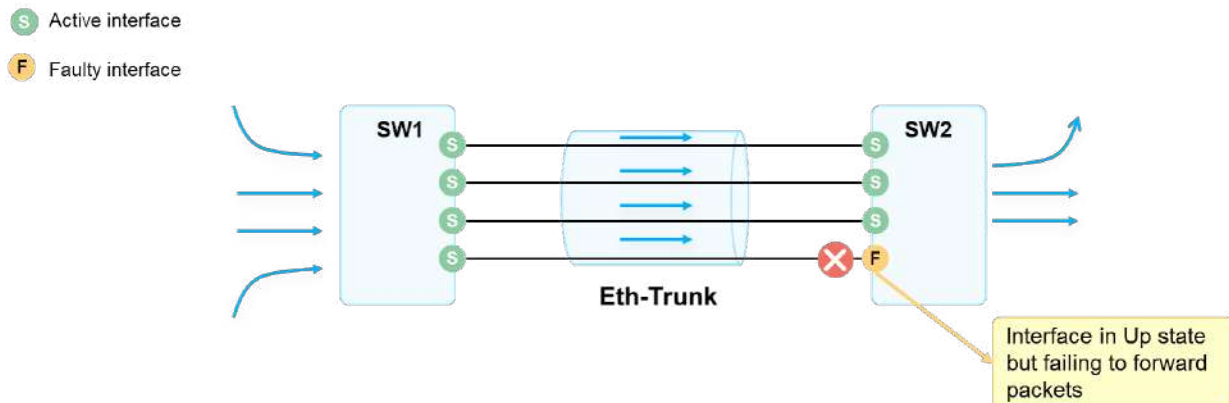


Manual aggregation mode

An Eth-Trunk is manually created, and its member interfaces are manually configured. LACP is not used for negotiation between the two systems. If an active link is faulty, the LAG automatically evenly shares traffic among the remaining active links.

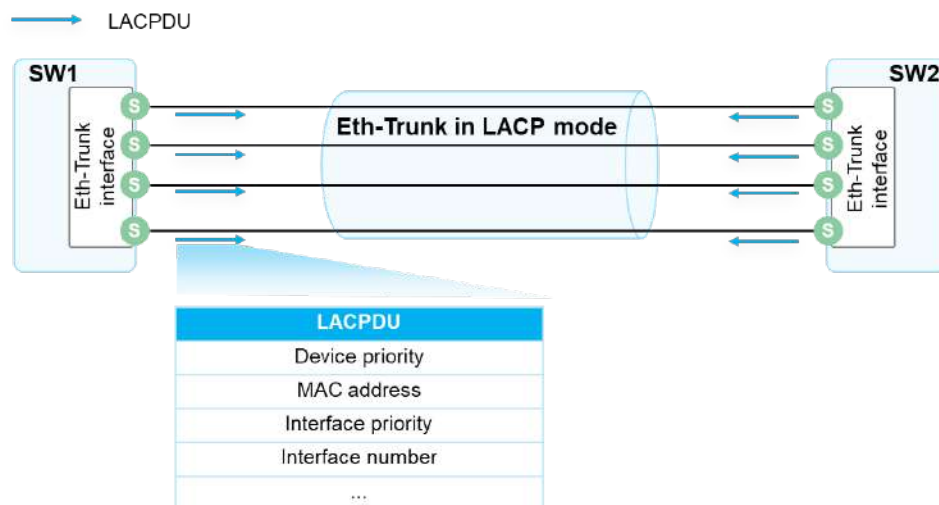


Following diagram illustrates the defect in Manual Mode



LACP aggregation mode

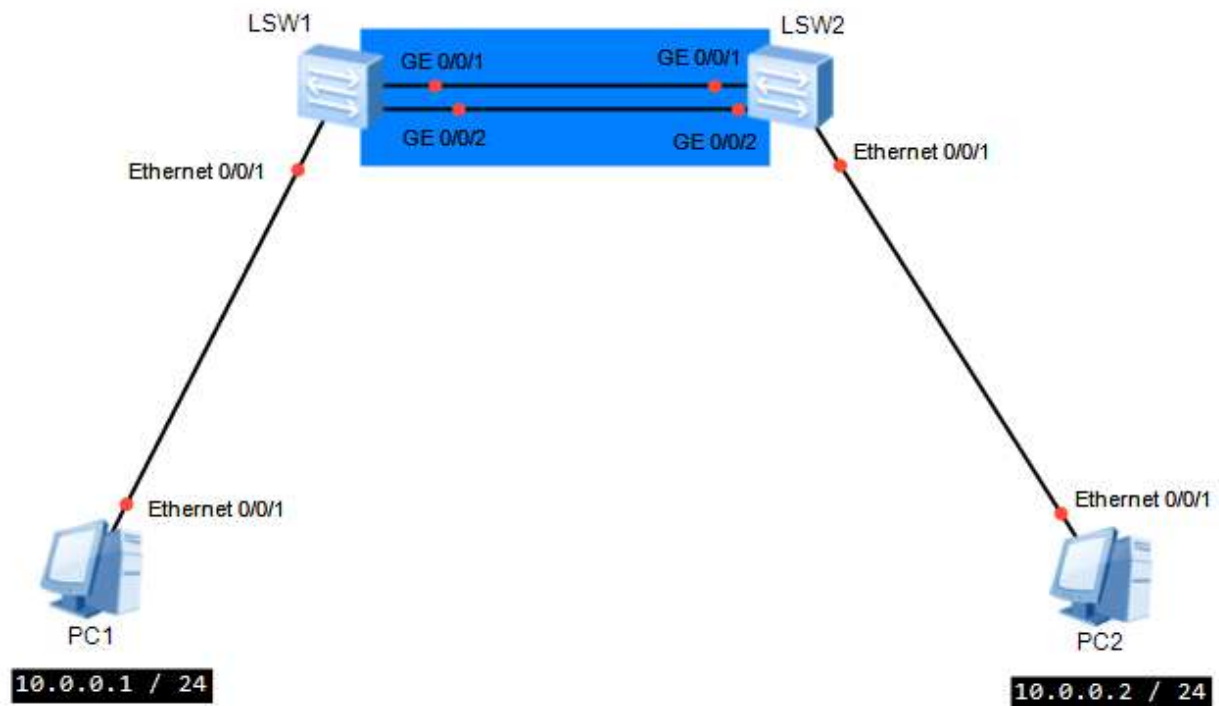
It is a link aggregation mode that uses the LACP protocol. Devices exchange Link Aggregation Control Protocol Data Units (LACPDU) to ensure that the peer interfaces are member interfaces that belong to the same Eth-Trunk and are on the same device.



By default, the LACP system priority is 32768. A smaller value indicates a higher priority. Generally, the default value is used. When the priorities are the same, LACP selects the Actor by comparing the MAC addresses. A smaller MAC address indicates a higher priority and a smaller interface number also indicates a higher priority.

Lab Task (Lab # 09):

Configure the following scenario by applying the LACP method (eth-trunk) in eNSP.



Lab # 10

Objective

Overview of Spanning Tree Protocol (STP)

Theory

Definition

Spanning Tree Protocol (STP), defined in IEEE 802.1D, builds a loop-free logical topology for LANs. It blocks redundant links between any two network devices and leaves a single active link between them so as to eliminate loops. If an active link fails, STP immediately selects a proper redundant link and places it in forwarding state to ensure network connectivity.

Why Do We Need STP?

Devices on a LAN are often interconnected using redundant links to improve service resilience. This configuration, however, is prone to loops. Loops can easily cause broadcast storms, which can consume sufficient network resources so as to break down the network. Loops can also cause MAC address flapping, damaging the MAC address entries.

This is where STP comes in. STP-capable devices exchange information with one another to discover loops on the network, and then block some redundant links to trim the ring network into a loop-free tree network. This prevents packets from being infinitely replicated or propagated on the network, and protects device performance from being deteriorated by repeated packets.



Figure 1. Need of STP to prevent loop

Important terms used in Spanning Tree Protocol

Here are some important terms used:

Bridge: Bridge is an important component of VTP, which connects two or more LAN segments.

Root Bridge (RB): It is the bridge that offers an interconnection point for all segments. All the bridges in a LAN have a path to the root. STP allows you to select the root bridge automatically. However, if the STP network admin wants, he or she can change the RB according to the network.

Non-Root Bridge (NRB): A non-root Bridge is any bridge that is not the root bridge.

Root Port (RP): The root port is a port that leads towards the Root Bridge.

Designated Port (DP): Every LAN segment has 1 Designated Port. Every bridge also receives the frames from DP and forwards them through its RP towards the Root Bridge.

Port ID: The port ID helps you to decide the root port. It consists of a configurable 1-byte priority value and a unique port number of each bridge.

Path Cost (PC): Path Cost, which is also known as PC, helps to decide the best topology with regard to forwarding speed. STP uses the concept of Path Cost.

Blocked Port: A non-Designated port is a switch port that is blocked. It is also known as a blocked port, or sometimes an alternate port. It cannot forward packets to the next switch. It also does not populate the MAC address in the MAC table.

How Does STP Work?

As shown in Figure 2, ports AP1 and AP2 reside on Device1, ports BP1 and BP2 reside on Device2, and ports CP1 and CP2 reside on Device3.

- Device1 sends configuration BPDUs to Device2 through AP1. Device1 is the designated bridge of Device2, and AP1 is the designated port on Device1.
- Device2 and Device3 are connected to the LAN. If Device2 sends configuration BPDUs to the LAN, it is the designated bridge of the LAN, and BP2 is the designated port on Device2.

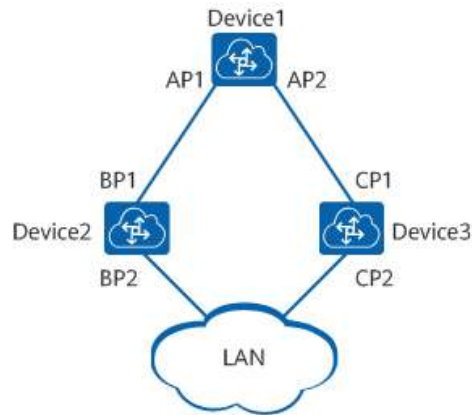


Figure 2. Designated bridge and designated port

In addition, a device has a bridge ID (BID) and a port on the device has a port ID (PID). A PID is composed of a port priority and a port number. A BID is composed of a bridge priority and a bridge MAC address. On an STP network, the device with the smallest BID is elected as the root bridge.

STP Implementation

STP is implemented as follows:

1. **Select the root bridge:** During network initialization, every STP-capable device considers itself the root bridge and sets the root bridge ID to its own BID. These devices then exchange configuration BPDUs and compare their root bridge IDs to find the one with the smallest root bridge ID, which will become the root bridge. All the ports on the root bridge are in Forwarding state.
2. **Select the root port:** A device that is not elected as the root bridge selects the port that receives the optimal configuration BPDU as the root port. The root port is in Forwarding state.
3. **Select the designated port:** The device calculates and generates a configuration BPDU for each port based on the configuration BPDU on the root port and path cost of the root port. The device then compares the generated configuration BPDU with the configuration BPDU received on the port.
 - If the generated configuration BPDU is superior, the port is selected as the designated port and periodically sends the generated configuration BPDU.
 - If the port's own configuration BPDU is superior, the port retains its own configuration BPDU and is blocked. In this case, the port only receives configuration BPDUs, and does not send them or forward data.

After the root bridge, root port, and designated ports are elected successfully, a tree topology is set up. When the topology is stable, only the root port and designated ports forward traffic. Other ports are in Blocking state; they only receive STP BPDUs and do not forward user traffic. The following example illustrates how STP calculation is implemented.

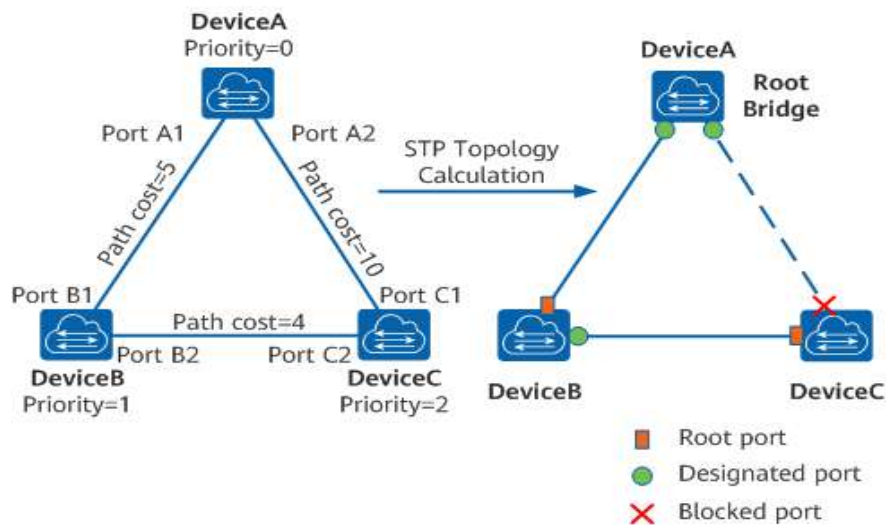


Figure 3. STP networking and calculated topology

In Figure 3, DeviceA, DeviceB, and DeviceC are deployed on the network, with priorities 0, 1, and 2, respectively. The path costs between DeviceA and DeviceB, DeviceA and DeviceC, and DeviceB and DeviceC are 5, 10, and 4, respectively.

Port State	Description
Disabled	The port cannot send or receive BPDUs or service data frames.
Blocking	A blocked port cannot send BPDUs but listens to BPDUs. In addition, the blocked port cannot send or receive service data frames or learn MAC addresses.
Listening	STP considers the port in the Listening state as the root port or designated port, but the port is still in the STP calculation process. In this case, the port can send and receive BPDUs but cannot send or receive service data frames or learn MAC addresses.
Learning	A port in Learning state listens to service data frames but cannot forward them. After receiving service data frames, the port learns MAC addresses.
Forwarding	A port in Forwarding state can send and receive service data frames and process BPDUs. Only the root port or designated port can enter the Forwarding state.

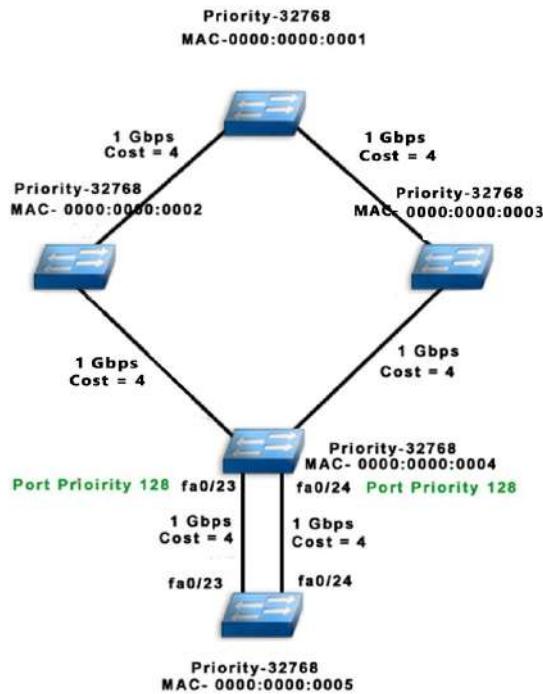
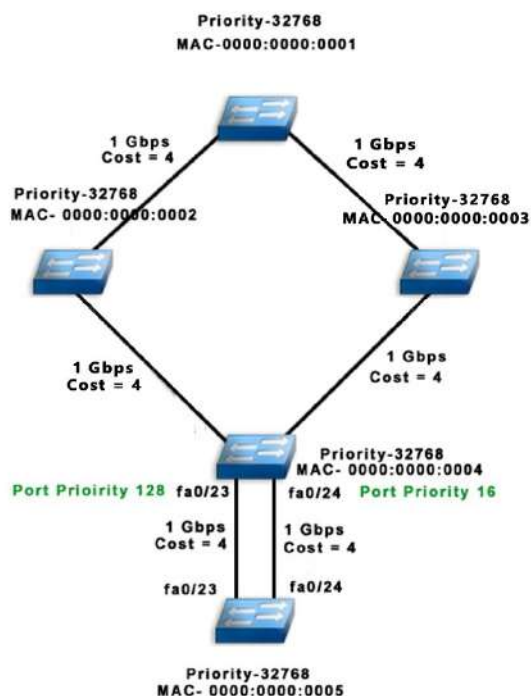
Table 1. Comparison among STP, RSTP, and MSTP

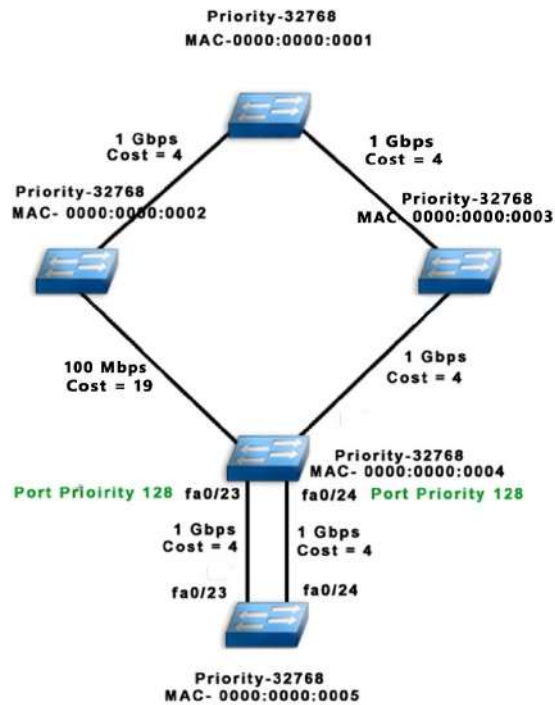
Spanning Tree Protocol	Characteristics	Application Scenario
STP	- Builds a loop-free tree topology to prevent broadcast storms and implement link redundancy. - Slow network topology convergence	All VLANs share one spanning tree without having to differentiate the traffic of users or services.
RSTP	- Builds a loop-free tree topology to prevent broadcast storms and implement link redundancy. - Fast network topology convergence	
MSTP	- Generates multiple loop-free trees to prevent broadcast storms and implement link redundancy. - Fast network topology convergence - Implements load balancing among VLANs, meaning that traffic in different VLANs is transmitted over different paths.	Traffic from different VLANs is forwarded through different spanning trees for load balancing. The spanning trees are independent of each other. In this situation, users or services are distinguished by VLANs.

Lab Task (Lab # 10):

State reason and identify the root bridge, root port (RP), designated port (DP) and alternative port (or blocked port), if STP is applied in the following scenarios:

NOTE: The last hextet number of each switch's MAC address is the switch number itself.

Scenario # 01**Scenario # 02**

Scenario # 03

Lab # 11

Objective

Overview of Dynamic Host Configuration Protocol (DHCP)

Theory

Definition

Dynamic Host Configuration Protocol is a network management protocol that is used to dynamically assign the IP address and other information to each host on the network so that they can communicate efficiently. DHCP automates and centrally manages the assignment of IP addresses, easing the work of network administrators. In addition to the IP address, the DHCP also assigns the subnet masks, default gateway and domain name server(DNS) address and other configuration to the host and by doing so, it makes the task of network administrator easier.

Components of DHCP

1. **DHCP Server:** It is typically a server or a router that holds the network configuration information.
2. **DHCP Client:** It is the endpoint that gets the configuration information from the server like any computer or mobile.
3. **DHCP Relay Agent:** If you have only one DHCP Server for multiple LAN's then the DHCP relay agent present in every network will forward the DHCP request to the servers. This is because the DHCP packets cannot travel across the router. Hence, the relay agent is required so that DHCP servers can handle the request from all the networks.
4. **IP address pool:** It contains the list of IP addresses which are available for assignment to the client.
5. **Subnet Mask:** It tells the host that in which network it is currently present.
6. **Lease Time:** It is the amount of time for which the IP address is available to the client. After this time the client must renew the IP address.
7. **Gateway Address:** The gateway address lets the host know where the gateway is to connect to the internet.

How does DHCP work?

DHCP works at the application layer to dynamically assign the IP address to the client and this happens through the exchange of a series of messages called DHCP transactions or DHCP conversation.

- **DHCP Discovery:** The DHCP client broadcasts messages to discover the DHCP servers. The client computer sends a packet with the default broadcast destination of 255.255.255.255 or the specific subnet broadcast address if any configured. 255.255.255.255 is a special broadcast address, which means “this network”: it lets you send a broadcast packet to the network you’re connected to.

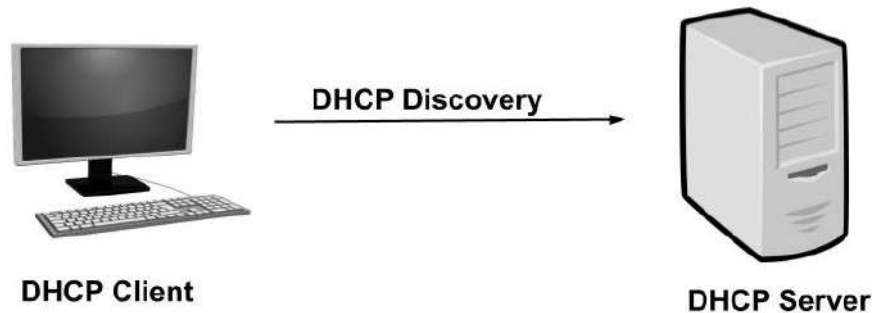


Figure 1. DHCP discovery

- **DHCP Offer:** When the DHCP server receives the DHCP Discover message then it suggests or offers an IP address (from IP address pool) to the client by sending a DHCP offer message to the client. This DHCP offer message contains the proposed IP address for DHCP client, IP address of the server, MAC address of the client, subnet mask, default gateway, DNS address, and lease information.

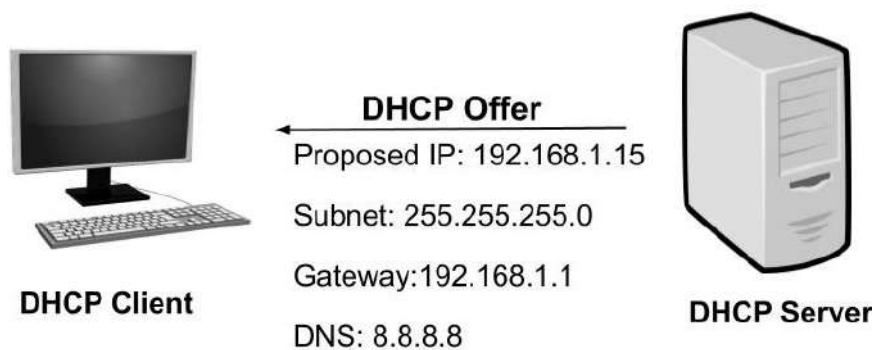


Figure 2. DHCP offer

1. the proposed IP address for DHCP client (here 192.168.1.11)
2. Subnet mask to identify the network (here 255.255.255.0)
3. IP of the default gateway for the subnet (here 192.168.1.1)
4. IP of DNS server for name translations (here 8.8.8.8)

- **DHCP Request:** In most cases, the client can receive multiple DHCP offers because in a network there are many DHCP servers(as they provide fault tolerance). If the IP addressing of one server fails then other servers can provide backup. But, the client will accept only one DHCP offer. In response to the offer, the client sends a DHCP Request requesting the offered address from one of the DHCP servers. All the other offered IP addresses from remaining DHCP servers are withdrawn and returned to the pool of IP available addresses.

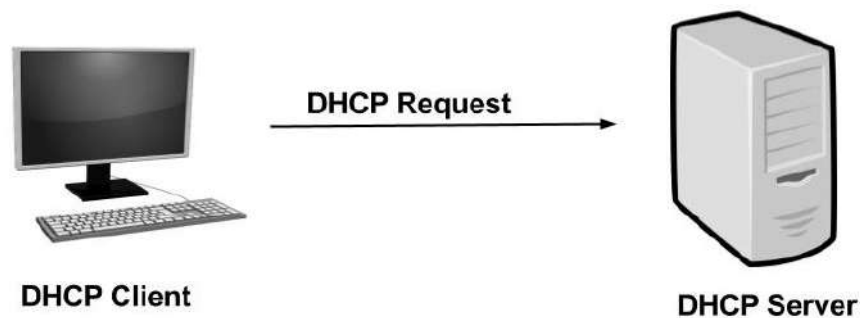


Figure 3. DHCP request

- **DHCP Acknowledgment:** The server then sends Acknowledgement to the client confirming the DHCP lease to the client. The server might send any other configuration that the client may have asked. At this step, the IP configuration is completed and the client can use the new IP settings.

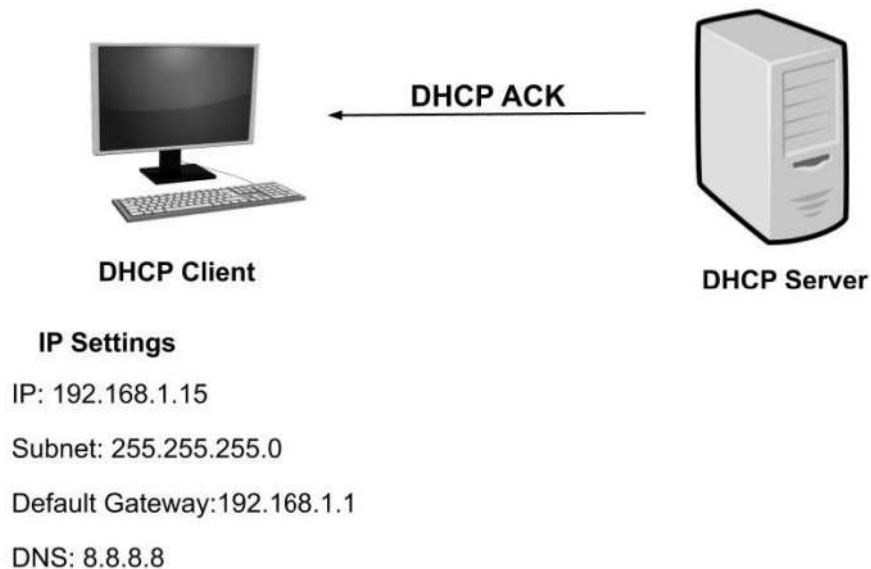
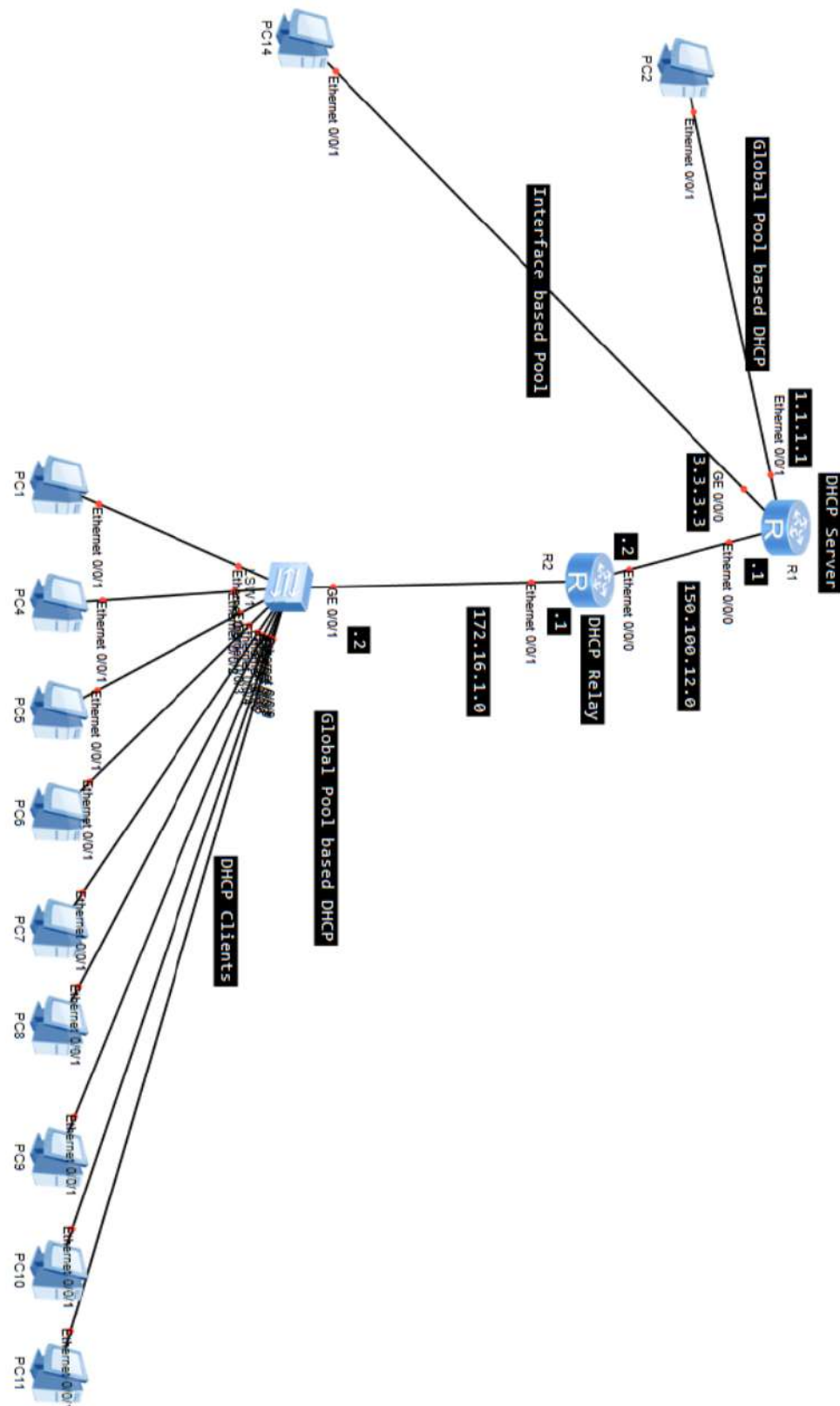


Figure 4. DHCP acknowledgement

Lab Task (Lab # 11):

Implement the following scenario by implementing the global pool and interface based DHCP server along with frame relay DHCP server.



Lab # 12

Objective

Dynamic Routing : Router Information Protocol (RIP)

Theory

Definition

Routing Information Protocol (RIP) is a dynamic routing protocol that uses hop count as a routing metric to find the best path between the source and the destination network. It is a distance-vector routing protocol. It is an Interior Gateway Protocol (IGP) designed to distribute routing information within an Autonomous System (AS).

Features

1. Updates of the network are exchanged periodically.
2. Updates (routing information) are always broadcast.
3. Full routing tables are sent in updates.
4. Routers always trust routing information received from neighbor routers. This is also known as Routing on rumors.

Versions

RIP v1 is known as Classful Routing Protocol because it doesn't send information about the subnet mask in its routing update.

RIP v2 is known as Classless Routing Protocol because it sends information about the subnet mask in its routing update.

RIP v1	RIP v2	RIPng
Sends update as broadcast	Sends update as multicast	Sends update as multicast
Broadcast at 255.255.255.255	Multicast at 224.0.0.9	Multicast at FF02::9 (RIPng can only run on IPv6 networks)
Doesn't support authentication of updated messages	Supports authentication of RIPv2 update messages	-
Classful routing protocol	Classless protocol updated supports classful	Classless updates are sent

RIP timers

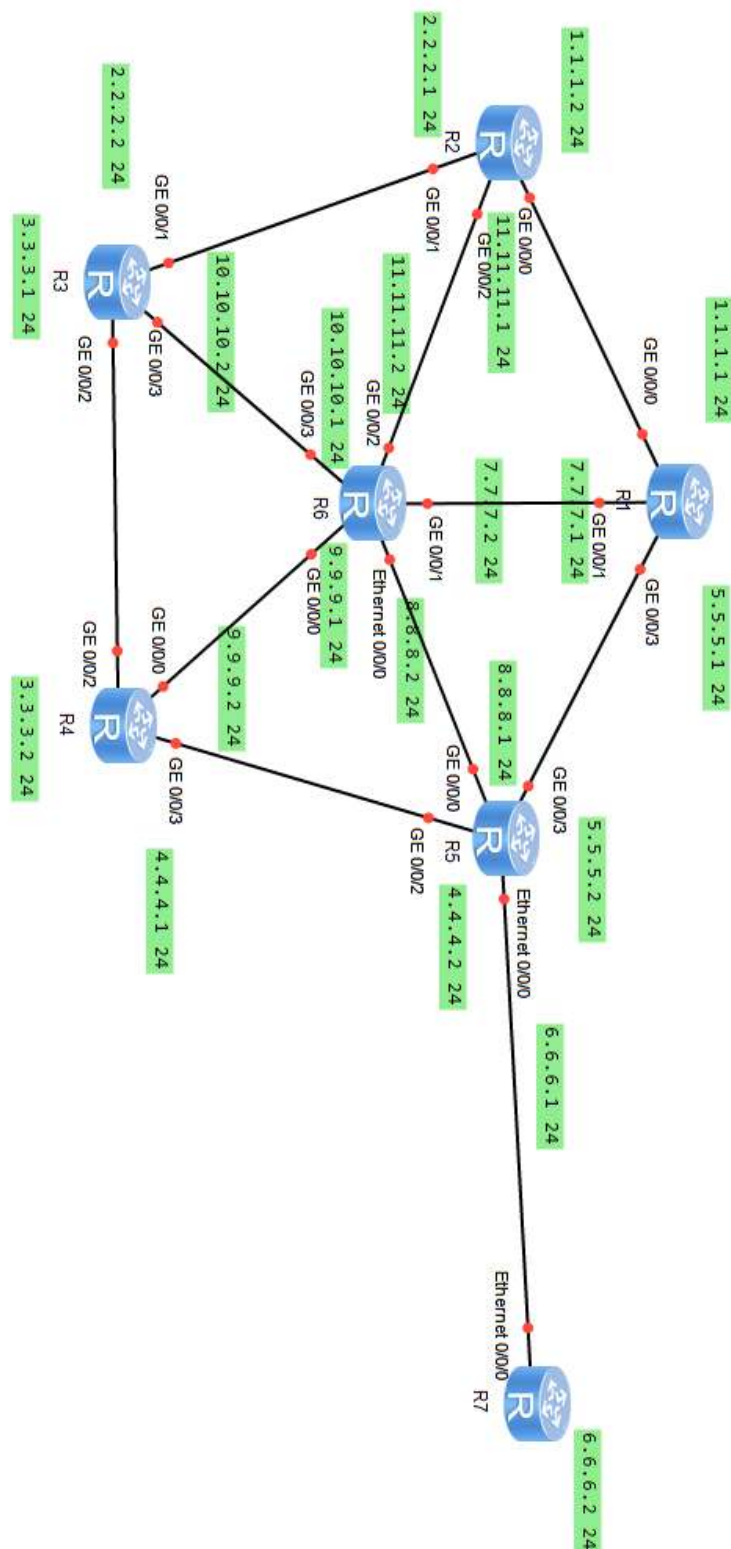
- **Update timer:** The default timing for routing information being exchanged by the routers operating RIP is 30 seconds. Using an Update timer, the routers exchange their routing table periodically.
- **Invalid timer:** If no update comes until 180 seconds, then the destination router considers it invalid. In this scenario, the destination router mark hop counts as 16 for that router.
- **Hold down timer:** This is the time for which the router waits for a neighbor router to respond. If the router isn't able to respond within a given time then it is declared dead. It is 180 seconds by default.
- **Flush time:** It is the time after which the entry of the route will be flushed if it doesn't respond within the flush time. It is 60 seconds by default. This timer starts after the route has been declared invalid and after 60 seconds i.e time will be $180 + 60 = 240$ seconds.

Working

- Each router initializes its routing table with a list of locally connected networks.
- Periodically, each router advertises the entire contents of its routing table over all of its RIP-enabled interfaces.
 - Whenever a RIP router receives such an advertisement, it puts all of the appropriate routes into its routing table and begins using it to forward packets. This process ensures that every network connected to every router eventually becomes known to all routers.
 - If a router does not continue to receive advertisements for a remote route, it eventually times out that route and stops forwarding packets over it. In other words, RIP is a "soft state" protocol.
- Every route has a property called a metric, which indicates the "distance" to the route's destination.
 - Every time a router receives a route advertisement, it increments the metric.
 - Routers prefer shorter routes to longer routes when deciding which of two versions of a route to program in the routing table.
 - The maximum metric permitted by RIP is 16, which means that a route is unreachable. This means that the protocol cannot scale to networks where there may be more than 15 hops to a given destination.

Lab Task (Lab # 12):

Implement the following scenario by applying RIP protocol (version 2).



Lab # 13

Objective

Console and Telnet Security

Theory

the PC and device are reachable to each other. Users require that the device be remotely configured and managed in an easy way. To meet the requirement, configure AAA authentication for Telnet users on the server and configure a security policy to allow only users meeting the policy to log in to the device.

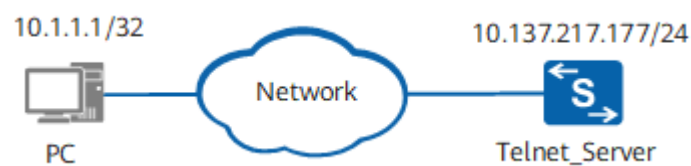


Figure 1. Networking diagram for configuring Telnet login

Configuration Roadmap

The configuration roadmap is as follows:

1. Log in to the device using Telnet to remotely maintain the device.
2. Configure the administrator user name and password, and configure an AAA authentication policy to ensure that only users passing the authentication can log in to the device.
3. Configure a security policy to ensure that only users meeting the policy can log in to the device.

Basic Concept of AAA (Authentication, authorization and accounting)

Authentication, authorization, and accounting (AAA) provides a management mechanism for network security.

A common AAA architecture includes the user, network access server (NAS), and AAA server

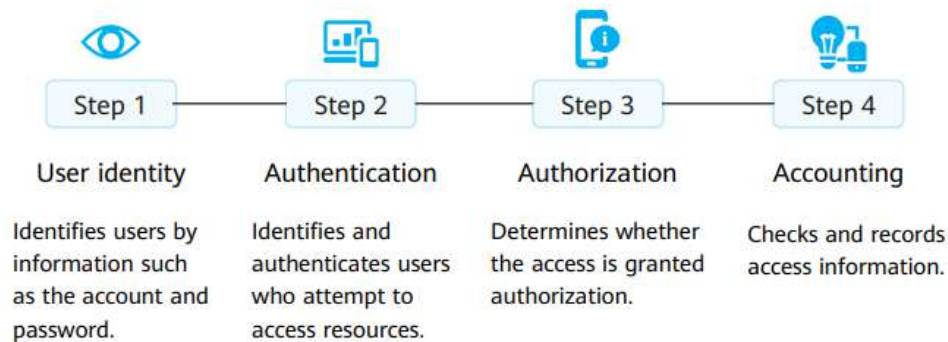


Figure 2. Basic Concept of AAA

AAA Configuration

1. Enter the AAA view.

```
[Huawei] aaa
```

Exit the system view and enter the AAA view.

2. Create an authentication scheme.

```
[Huawei-aaa] authentication-scheme authentication-scheme-name
```

Create an authentication scheme and enter the authentication scheme view.

```
[Huawei-aaa-authentication-scheme-name] authentication-mode { hwtacacs | local | radius }
```

Set the authentication mode to local authentication. By default, the authentication mode is local authentication.

3. Create a domain and bind an authentication scheme to the domain.

```
[Huawei-aaa] domain domain-name
```

Create a domain and enter the domain view

```
[Huawei-aaa-domain-name] authentication-scheme authentication-scheme-name
```

Bind the authentication scheme to the domain

4. Create a user.

[Huawei-aaa] **local-user** *user-name* **password cipher** *password*

Create a local user and configure a password for the local user

- If the username contains a delimiter "@", the character before "@" is the username and the character after "@" is the domain name.
- If the value does not contain "@", the entire character string represents the username and the domain name is the default one.

5. Configure a user access type.

[Huawei-aaa] **local-user** *user-name* **service-type** { { **terminal** | **telnet** | **ftp** | **ssh** | **snmp** | **http** } | **ppp** | **none** }

Configure the access type of the local user. By default, all access types are disabled for a local user

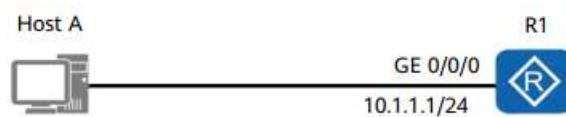
6. Configure a user level.

[Huawei-aaa] **local-user** *user-name* **privilege level** *level*

Specify the permission level of the local user.

AAA Configuration Example

After a user password and a user level are configured on R1, host A can use the configured username and password to remotely log in to R1.



```
[R1]aaa
[R1-aaa]local-user huawei password cipher huawei123
[R1-aaa]local-user huawei service-type telnet
[R1-aaa]local-user huawei privilege level 0
[R1]user-interface vty 0 4
[R1-ui-vty0-4]authentication-mode aaa
```

Figure 2. AAA Configuration Example

Lab Task (Lab # 13):

Add console and vty security access settings in each router of Lab 12.

Lab # 14

Objective

Dynamic Routing : Open Shortest Path First (OSPF)

Theory

Open Shortest Path First (OSPF) is a link-state routing protocol that is used to find the best path between the source and the destination router using its own Shortest Path First). It is a network layer protocol.

OSPF terms

1. Router Id – It is the highest active IP address present on the router. First, the highest loopback address is considered. If no loopback is configured then the highest active IP address on the interface of the router is considered.
2. Router priority – It is an 8-bit value assigned to a router operating OSPF, used to elect DR and BDR in a broadcast network.
3. Designated Router (DR) – It is elected to minimize the number of adjacencies formed. DR distributes the LSAs to all the other routers. DR is elected in a broadcast network to which all the other routers share their DBD. In a broadcast network, the router requests for an update to DR, and DR will respond to that request with an update.

Backup Designated Router (BDR) – BDR is a backup to DR in a broadcast network. When DR goes down, BDR becomes DR and performs its functions.

Area 0

This is the backbone area of an OSPF routing domain. An OSPF routing domain (AS) can consist of a single area, but in that case the single area must be Area 0.

Those routers which are in Area 0 and all their interfaces are in Area 0, such routers are called backbone routers.

Area Border Router (ABR)

A Router is said to be an area border router if its minimum one interface is in OSPF AREA 0 and other interfaces can be in any area (Area 1 – Area $2^{32}-1$).

Autonomous System Boundary Router (ASBR)

A Router is said to be ASBR if its one interface is in OSPF Domain and Other Interface in any Other Routing Protocol Domain (RIP , EIGRP) . But we have to redistribute in order to make Router R3 as ASBR, such as RIP etc are ASBR

Designated Router (DR):

A Designated Router is elected by the routers on multi-access segments (e.g Local Area Network), based on its priority (Router ID, priority). The DR router performs special functions such as generating Link State Advertisements (LSAs) and exchanging information with all other routers in the same Area.

Backup Designated Router (BDR):

The BDR is a router that becomes the DR should the existing DR fail. The BDR has the second highest priority (the DR having the highest priority) in the OSPF network. When the BDR becomes a DR, a new election is held to find a new BDR.

Steps to Select Router Id

1. Manual configuration of the router ID.
2. Highest IP address on a loopback interface.
3. Highest IP address on a non-loopback interface.

DR and BDR election

DR and BDR election takes place in the broadcast network or multi-access network. Here are the criteria for the election:

1. Routers having the highest router priority will be declared as DR.
2. If there is a tie in router priority then the highest router ID be considered. First, the highest loopback address is considered. If no loopback is configured then the highest active IP address on the interface of the router is considered.

Lab Task (Lab # 14):

Implement multiple OSPF areas with RIP protocol in eNSP using following scenario:

